

Behavioral UML Documentation and Hyperledger Fabric Mapping of Procurement Business Processes for an E-Procurement Service

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Abstract—Procurement work is often executed through fragmented approvals, email-based quotations, manual purchase-order preparation, paper delivery confirmations, spreadsheet-based evaluations, and accounts-payable exception handling. These handoffs make it difficult for engineers to determine which process flows should be digitized, automated, or supported by tamper-evident records. This paper presents an IEEE-style behavioral modeling study of procurement business processes for an e-procurement service integrated with Hyperledger Fabric. The study identifies actors, terminology, process objectives, triggers, inputs, outputs, normal flows, alternative flows, exception flows, business rules, inefficiencies, and streamlining opportunities. It then maps each process to frontend interfaces, backend services, database entities, blockchain assets, Fabric chaincode functions, and ledger events. The paper provides a UML-oriented documentation package using use case, activity, sequence, state-machine, and component-level representations. The analysis argues that blockchain is not a universal replacement for procurement databases; rather, it is most defensible for multi-party trust, non-repudiation, traceability, and auditability of critical procurement milestones. Sensitive documents and detailed commercial data should remain off-chain, while ledger entries should store lifecycle states, hashes, identities, timestamps, and events.

Index Terms—e-procurement, business process analysis, UML, activity diagram, sequence diagram, Hyperledger Fabric, blockchain, chaincode, procurement audit, procure-to-pay, source-to-contract

I. INTRODUCTION

Procurement is a cross-functional business capability that connects demand identification, supplier selection, ordering, receiving, invoicing, payment, compliance, and supplier performance management. In many organizations, the process is only partially digitized: requests are created in email, approvals occur through informal messages, quotation comparisons are performed in spreadsheets, purchase orders are rekeyed into finance systems, and invoice matching requires manual reconciliation. These practices increase cycle time, data inconsistency, maverick spending, contract leakage, invoice exceptions, and audit cost. Prior procurement research and benchmark evidence show that process redesign, automation, data quality, system compliance, and supplier adoption strongly influence e-procurement value [11]–[15].

The research task addressed in this paper is: how can selected procurement business processes be documented, visualized, and analyzed using behavioral UML diagrams so that engineers can understand current process flow, iden-

tify inefficiencies, and map each process to a possible e-procurement service architecture supported by frontend interfaces, backend services, databases, and Hyperledger Fabric blockchain networks? The contribution is a structured process catalog, use case specification, UML diagram inventory, and architecture mapping that future software engineers can use as a requirements baseline.

This paper is not a full software implementation specification. It does not prescribe a specific ERP product, cloud vendor, programming language, database engine, or legal compliance regime. It models a generic medium-to-large buying organization. Organization-specific facts, thresholds, supplier categories, approval limits, tax rules, and payment regulations must be configured during implementation.

II. TASK INTERPRETATION, SCOPE, AND ASSUMPTIONS

The task is interpreted as a business-process and system-design study rather than a code build. The selected processes span source-to-contract (S2C), procure-to-pay (P2P), supplier management, accounts payable, compliance, and exception handling. The design objective is educational: business readers should understand process meaning and engineering readers should see how process events become system services, data records, and Fabric ledger events.

The assumed process scope includes the fourteen processes listed in the prompt and one additional process: supplier performance and contract closeout. It is included because procurement does not end at payment; engineers need feedback loops from delivery quality, disputes, invoice exceptions, and contract closeout to future supplier selection. The main assumptions are: (i) the buyer organization controls the e-procurement application; (ii) suppliers access the system through a portal; (iii) an operational database remains the source for mutable application data and large documents; (iv) Hyperledger Fabric stores asset lifecycle events, hashes, selected non-sensitive metadata, and audit states; and (v) private data collections may be used where subsets of organizations need restricted ledger-side data sharing.

III. BACKGROUND AND RELATED CONCEPTS

A. Procurement, S2C, and P2P

Procurement is the organized acquisition of goods and services needed by an organization. In a systems context,

S2C includes requirement definition, market analysis, RFQ or tendering, supplier evaluation, negotiation, award, and contract creation. P2P includes purchase requisition, approval, purchase order, goods or service receipt, invoice verification, and payment. A P2P definition normally treats requisition, sourcing or ordering, receiving, invoicing, and payment as one connected flow [3], [4].

B. Behavioral UML for Process Understanding

UML is a standard graphical language for visualizing and documenting software-intensive systems [1]. This paper emphasizes behavioral diagrams because procurement is primarily a workflow problem: use case diagrams identify externally visible goals; activity diagrams show control flow and decisions; sequence diagrams show messages among actors and system components; and state-machine diagrams show lifecycle states of business assets such as VendorProfile, RFQ, PurchaseOrder, Invoice, and PaymentApproval.

C. Hyperledger Fabric as Procurement Audit Infrastructure

Hyperledger Fabric is an enterprise-oriented permissioned distributed ledger platform [5]. Fabric peers host ledgers and chaincode, client applications submit transaction proposals, endorsement policies specify which organizations must endorse, and the ordering service orders transactions into blocks for peer validation and commit [6], [8]. The Membership Service Provider (MSP) and certificate authorities establish trusted identities for organizations and users [7], [9]. Fabric private data collections can keep sensitive data visible only to an authorized subset of organizations while still anchoring hashes on the channel ledger [10].

IV. METHODOLOGY

The method follows eight steps. First, procurement processes are identified from the domain scope and from common S2C/P2P practice. Second, actors and stakeholders are defined. Third, a business glossary normalizes terms so that analysts and engineers use the same vocabulary. Fourth, each process is decomposed into objective, trigger, actors, preconditions, inputs, outputs, normal flow, alternative flow, exception flow, rules, pain points, e-procurement opportunity, and blockchain opportunity. Fifth, process outputs are mapped to system components: frontend screens, backend services, database entities, Fabric transactions, chaincode functions, and ledger events. Sixth, UML behavioral representations are prepared as PlantUML source. Seventh, current manual or semi-manual processes are compared with a digitized process supported by workflow automation and ledger-backed audit. Finally, limitations and recommendations are derived.

V. OVERALL BUSINESS PROCESS CONTEXT

A. Actor and Stakeholder Analysis

Table II lists the main actors. The architecture distinguishes human actors from system components. A supplier, for example, is a business actor, while the supplier portal is a frontend

interface. Similarly, Hyperledger Fabric is not a human stakeholder but a network infrastructure that enforces identity-aware transaction validation and audit-state replication.

B. Simplified Actor-System Interaction Model

Fig. 1 provides a simplified architecture view. It is intentionally high-level because the purpose is to make process responsibilities visible before implementation details are selected.

VI. BUSINESS PROCESS CATALOG

Table III catalogs the processes in scope. Blockchain relevance is intentionally qualified. A high blockchain relevance does not mean all data should be on-chain; it means the process has valuable audit, non-repudiation, cross-party, or lifecycle-state evidence.

VII. USE CASE SPECIFICATIONS

Table VI summarizes the use case specifications. The table is compressed for IEEE layout; the narrative subsections that follow provide detailed process interpretation.

VIII. DETAILED BUSINESS PROCESS ANALYSIS

This section records each business process in a repeatable analytical structure. The repeated structure helps engineers trace business requirements into workflow screens, backend APIs, database entities, and Fabric transactions.

1) *BP01: Vendor Registration and Onboarding: Objective.* Create a trusted supplier record before any sourcing, purchase order, invoice, or payment activity is permitted. *Trigger.* New supplier wishes to transact or is invited by procurement *Actors*. The primary actor is Vendor/Supplier; supporting actors are Procurement Officer, Compliance Officer, System Administrator. *Preconditions.* Supplier is eligible to apply; onboarding policy, required documents, tax identifiers, and bank validation rules are defined. *Inputs and outputs.* Inputs include Company profile, registration number, tax data, bank details, certifications, contact persons, beneficial ownership declarations, risk questionnaires.. Outputs include VendorProfile record, onboarding status, approval decision, compliance evidence hash, VendorRegistered or VendorRejected event. *Normal flow.* 1) Supplier completes self-service registration form and uploads evidence. 2) Backend validates mandatory fields, duplicates, document format, and risk-screening requirements. 3) Procurement and compliance reviewers approve or reject the supplier. 4) Database stores full documents off-chain; chaincode records profile hash, status, reviewer identity, and timestamp. 5) System notifies supplier and procurement users. *Alternative and exception flows.* Alternative: If documents are incomplete, the application returns to the supplier for correction without creating an active supplier account. Exception: If fraud, sanctions, duplicate bank account, or identity mismatch is detected, the case is blocked and escalated to compliance. *Business rules.* No RFQ, PO, contract, invoice, or payment can reference an unapproved supplier; sensitive bank documents remain off-chain. *Inefficiency and improvement.* Current pain

TABLE I
GLOSSARY OF BUSINESS AND BLOCKCHAIN TERMS

Term	Explanation for business and engineering audiences
Procurement	The organizational process of obtaining goods and services needed to operate, including sourcing, ordering, receiving, invoicing, and supplier management.
Purchase requisition	An internal request to buy goods or services before a purchase order is issued; it captures need, quantity, budget, and justification.
Purchase order	A formal buyer document sent to a supplier specifying items, quantities, prices, delivery terms, and payment terms.
Request for quotation (RFQ)	A sourcing document asking suppliers to submit price and delivery offers for defined requirements.
Supplier onboarding	The process of collecting, verifying, and approving supplier identity, tax, bank, compliance, and risk data.
Approval workflow	A rule-based sequence of authorization tasks that ensures spending is approved by the correct roles and thresholds.
Three-way matching	The control that compares purchase order, goods receipt or service confirmation, and invoice before payment.
Invoice verification	The accounts-payable process of validating invoice data, tax details, supplier identity, duplicates, and matching results.
Audit trail	A chronological evidence record showing who performed each action, when, and with what data or decision.
Smart contract	Executable business logic on a blockchain. In Hyperledger Fabric, smart contracts are implemented as chaincode.
Chaincode	Fabric application code that validates transactions and reads or writes ledger state.
Ledger	The shared record of validated transactions and current asset state maintained by Fabric peers on a channel.
Endorsement policy	A Fabric rule defining which organizations or peers must endorse a transaction before it can be ordered and committed.
Ordering service	The Fabric service that orders endorsed transactions into blocks and distributes them to peers for validation and commit.
Peer node	A Fabric node that hosts ledgers and chaincode, endorses transaction proposals, and commits valid blocks.
Certificate Authority	The authority that issues cryptographic identities used by participants in the permissioned network.
Membership Service Provider (MSP)	The Fabric mechanism that defines trusted identities, roles, and organizations allowed to participate in channels and policies.
Immutability	The property that committed ledger entries are append-only and tamper-evident after validation.
Non-repudiation	A control objective where a participant cannot plausibly deny a signed action or submitted transaction.
Traceability	The ability to follow the lifecycle of procurement assets across requisition, RFQ, quotation, PO, delivery, invoice, payment, and audit.

TABLE II
ACTORS AND STAKEHOLDERS

Actor	Business role	System permissions	Typical data handled
Requesting Department/User	Identifies the business need, creates requisitions, confirms services, and supplies specifications.	Create requisition; view status; provide clarifications; confirm service acceptance.	Need description, quantity, cost center, justification, service acceptance.
Approver/Manager	Authorizes spending and exceptions based on budget, authority, and policy.	Approve/reject/return requests; delegate authority; approve exceptions.	Approval decision, comments, timestamp, delegation evidence.
Procurement Officer	Runs sourcing, supplier selection, RFQ, PO, contract, and supplier performance activities.	Create RFQs, evaluate bids, issue POs, manage contracts and exceptions.	RFQ data, award recommendation, PO, contract terms, case decisions.
Vendor/Supplier	Registers profile, submits quotations, acknowledges POs, delivers goods/services, and submits invoices.	Self-service onboarding; bid submission; PO acknowledgement; invoice status.	Company data, quotation, delivery note, invoice, dispute evidence.
Finance/AP Officer	Verifies invoices, matches PO/receipt/invoice, approves payments, and maintains payment controls.	Invoice verification, payment approval, exception routing.	Invoice status, match result, payment approval, bank validation reference.
Receiving Officer/Service Owner	Confirms delivery quantity, quality, and service completion.	Record goods receipt; accept/reject delivery; upload evidence.	Goods receipt, inspection result, photos, service acceptance.
Auditor/Compliance Officer	Reviews policy compliance, risk, audit evidence, and remediation.	Read-only evidence packs; create findings; close remediation.	Audit findings, exception classification, control test evidence.
System Administrator	Manages roles, workflow rules, integration configuration, and access controls.	Maintain users/roles/rules; configure APIs and Fabric connection.	User-role mapping, approval matrix, channel policy configuration.
Hyperledger Fabric Network	Provides permissioned ledger, chaincode, endorsement, ordering, and immutable event history.	Endorse and commit transactions; emit events; enforce asset lifecycle rules.	Asset state, transaction log, event stream, private-data hash.

point: Manual onboarding causes duplicate supplier records, inconsistent compliance checks, email-based document loss, and weak evidence for later audit. E-procurement opportunity: Supplier portal, workflow routing, automatic completeness checks, risk APIs, and master-data governance reduce cycle time and rework. Blockchain opportunity: Fabric provides non-repudiable onboarding events, status history, identity-based access control, and tamper-evident evidence hashes.

2) *BP02: Purchase Requisition Creation: Objective.* Capture the business need in a structured request before sourcing or purchasing begins. *Trigger.* Business need for goods or services *Actors.* The primary actor is Requesting Department;

supporting actors are Procurement Officer, Budget Owner. *Preconditions.* Requester is authenticated; item catalogs, budget codes, approval thresholds, and procurement policies are configured. *Inputs and outputs.* Inputs include Item or service description, quantity, estimated amount, required date, cost center, justification, preferred supplier where allowed.. Outputs include PurchaseRequisition record, validation result, RequisitionCreated event, approval workflow task. *Normal flow.* 1) Requester searches catalog or creates a non-catalog request. 2) Frontend validates required fields and budget code format. 3) Backend calculates approval route and stores requisition draft or submitted record. 4) Chaincode records a digest of the

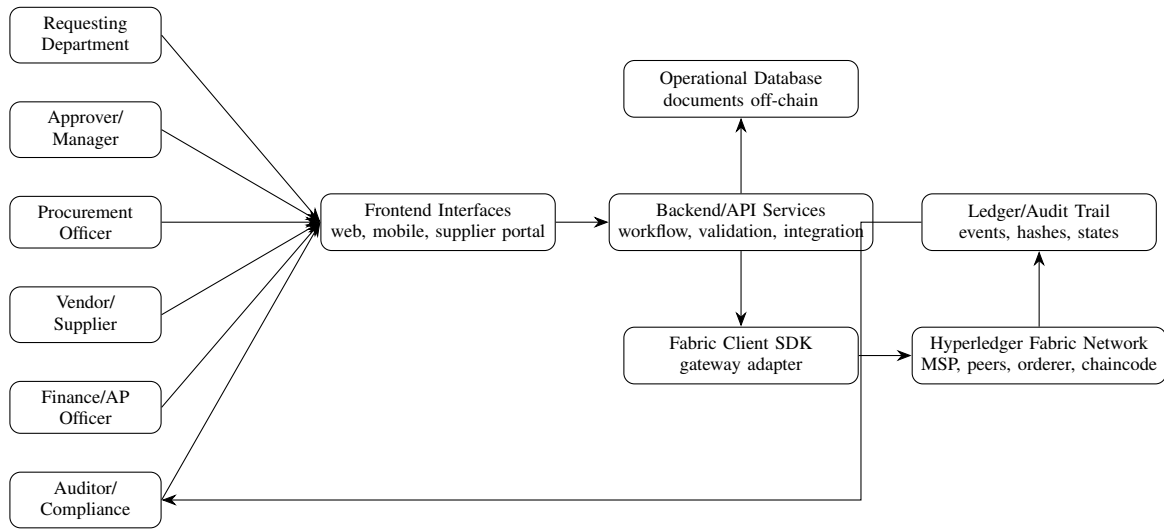


Fig. 1. Simplified actor-system interaction model. Human actors use frontend interfaces; backend services validate workflow and persist operational records; Fabric records selected procurement states, hashes, and events for auditability.

TABLE III
BUSINESS PROCESS CATALOG (PART 1 OF 3)

Process ID	Business Process Name	Primary Actor	Supporting Actors	Trigger	Main Output	Blockchain Relevance	Streamlining Opportunity
BP01	Vendor Registration and Onboarding	Vendor/Supplier	Procurement Officer, Compliance Officer, System Administrator	New supplier wishes to transact or is invited by procurement	Approved VendorProfile or rejection notice	High: identity, compliance status, bank-data attestation hash, immutable onboarding history	Supplier self-service, automated validation, risk scoring, duplicate detection
BP02	Purchase Requisition Creation	Requesting Department	Procurement Officer, Budget Owner	Business need for goods or services	Submitted PurchaseRequisition	Medium: hash of submitted requisition, requester identity, budget category, version history	Guided buying, catalog selection, budget validation, form standardization
BP03	Purchase Request Approval	Approver/Manager	Requesting Department, Procurement Officer, Finance Officer	Submitted requisition requires approval	Approved or rejected requisition	High: approval decision, digital signature, authority level, timestamp	Rule-based routing, escalation, delegation, threshold controls
BP04	Request for Quotation Preparation and Publication	Procurement Officer	Requesting Department, Supplier, Approver	Approved requisition requires competitive quotation or tendering	Published RFQ/RFP/Tender package	High: RFQ publication time, eligible suppliers, immutable terms hash	Template reuse, supplier invitation, electronic bid windows, evaluation rules
BP05	Supplier Quotation Submission	Vendor/Supplier	Procurement Officer, System Administrator	Supplier receives RFQ invitation or public tender access	Submitted Quotation	High: bid timestamp, confidentiality hash, supplier identity, anti-tampering evidence	Electronic bid submission, deadline enforcement, sealed-bid controls

TABLE IV
BUSINESS PROCESS CATALOG (PART 2 OF 3)

Process ID	Business Process Name	Primary Actor	Supporting Actors	Trigger	Main Output	Blockchain Relevance	Streamlining Opportunity
BP06	Quotation Evaluation and Supplier Award	Procurement Officer	Evaluation Committee, Requesting Department, Approver, Auditor	RFQ closes and quotations are available for evaluation	Award recommendation and selected supplier	High: evaluation scoring hash, award decision, committee approvals, conflict-of-interest evidence	Evaluation templates, scoring matrix, comparative analytics, approval routing
BP07	Purchase Order Generation and Supplier Acceptance	Procurement Officer	Supplier, Finance Officer, Requesting Department	Approved award, catalog requisition, or contract release	Issued and acknowledged PurchaseOrder	High: PO commitment, supplier acknowledgement, version history	Auto-generation from requisition/award, electronic dispatch, acknowledgement tracking
BP08	Contract Creation and Approval	Procurement Officer	Legal Officer, Supplier, Approver, Compliance Officer	Award requires formal contract or framework agreement	Approved and active Contract	High: contract hash, approval signatures, version and amendment trail	Template clauses, approval workflow, obligation extraction, renewal alerts
BP09	Goods or Service Delivery Confirmation	Receiving Officer	Supplier, Requesting Department, Procurement Officer	Supplier delivers goods or completes service milestone	GoodsReceipt or ServiceAcceptance record	High: delivery confirmation, acceptance/rejection, quantity and timestamp evidence	Mobile receiving, barcode/QR capture, automated PO matching
BP10	Invoice Submission	Vendor/Supplier	Finance Officer, Procurement Officer	Supplier has delivered goods/services and seeks payment	Submitted Invoice	Medium-High: invoice hash, submission timestamp, supplier identity, duplicate prevention	E-invoicing, PO flip, mandatory validation, invoice status self-service

TABLE V
BUSINESS PROCESS CATALOG (PART 3 OF 3)

Process ID	Business Process Name	Primary Actor	Supporting Actors	Trigger	Main Output	Blockchain Relevance	Streamlining Opportunity
BP11	Invoice Verification and Three-Way Matching	Finance Officer	Procurement Officer, Receiving Officer, Supplier	Invoice is submitted and ready for validation	Matched invoice, exception case, or rejection	High: match result, tolerance decision, exception resolution history	Automated two-way/three-way matching, tolerance rules, exception queues
BP12	Payment Approval and Payment Instruction	Finance Officer	Approver/Manager, Supplier, Treasury/Bank Interface	Invoice is matched or exception approval is complete	Approved payment instruction and payment status	Medium-High: approval timestamp, payer identity, payment authorization hash	Payment approval queues, segregation checks, bank integration, supplier status visibility
BP13	Procurement Audit and Compliance Review	Auditor/Compliance Officer	Procurement Officer, Finance Officer, System Administrator	Scheduled audit, risk alert, dispute, management review, or regulatory request	Audit findings, compliance status, remediation actions	Very high: immutable process chronology, evidence hashes, chain of custody	Digital audit trails, automated exception reports, evidence dashboards
BP14	Dispute or Exception Handling	Procurement Officer	Supplier, Finance Officer, Receiving Officer, Auditor, Approver	Mismatch, rejected delivery, supplier protest, invoice exception, contract dispute, or policy violation	Resolved exception, corrective action, or escalated dispute	High: dispute timeline, evidence hashes, decisions, settlement record	Exception case management, SLA tracking, root-cause analytics
BP15	Supplier Performance and Contract Closeout	Procurement Officer	Requesting Department, Supplier, Compliance Officer, Auditor	Delivery completion, contract milestone, periodic review, or contract expiry	Supplier scorecard, closeout record, renewal/termination decision	Medium-High: performance score hash, milestone completion, closeout evidence	Automated scorecards, KPI tracking, renewal alerts, performance feedback loops

TABLE VI
USE CASE SPECIFICATION SUMMARY (PART 1 OF 5)

Use Case ID	Use Case Name	Actor	Goal	Preconditions	Main Flow	Alternative Flow	Exception Flow	Postconditions
UC01	Vendor Registration and Onboarding	Vendor/Supplier	Create a trusted supplier record before any sourcing, purchase order, invoice, or payment activity is permitted.	Supplier is eligible to apply; onboarding policy, required documents, tax identifiers, and bank validation rules are defined.	Supplier completes self-service registration form and uploads evidence.; Backend validates mandatory fields, duplicates, document format, and risk-screening requirements.; Procurement and compliance reviewers approve or reject the supplier.	If documents are incomplete, the application returns to the supplier for correction without creating an active supplier account.	If fraud, sanctions, duplicate bank account, or identity mismatch is detected, the case is blocked and escalated to compliance.	VendorProfile record, onboarding status, approval decision, compliance evidence hash, VendorRegistered or VendorRejected event.
UC02	Purchase Requisition Creation	Requesting Department	Capture the business need in a structured request before sourcing or purchasing begins.	Requester is authenticated; item catalogs, budget codes, approval thresholds, and procurement policies are configured.	Requester searches catalog or creates a non-catalog request.; Frontend validates required fields and budget code format.; Backend calculates approval route and stores requisition draft or submitted record.	If the item is in a managed catalog, the system pre-populates supplier, price, category, and contract reference.	If budget is exhausted or required data are missing, the requisition remains in draft or is returned for revision.	PurchaseRequisition record, validation result, RequisitionCreated event, approval workflow task.
UC03	Purchase Request Approval	Approver/Manager	Authorize spending before a sourcing event or purchase order is executed.	Requisition is submitted; approval matrix, delegation rules, and budget checks are configured.	Backend routes the request to the correct approver by cost center, category, and value.; Approver reviews amount, justification, budget, and policy warnings.; Approver approves, rejects, or requests clarification.	If the approver delegates authority, the delegated approver acts under an auditable delegation rule.	If approval exceeds defined authority or segregation-of-duties rules fail, the workflow is blocked and escalated.	Approval record, approved/rejected status, audit entry, next workflow task.

submitted requisition and initial lifecycle state. 5) Approvers are notified. *Alternative and exception flows.* Alternative: If the item is in a managed catalog, the system pre-populates supplier, price, category, and contract reference. Exception: If budget is exhausted or required data are missing, the requisition remains in draft or is returned for revision. *Business rules.* Requisitions above threshold require managerial approval; direct PO creation is not allowed unless policy permits blanket or catalog buying. *Inefficiency and improvement.* Current pain point: Email or spreadsheet requisitions are incomplete, hard to track, and often require rekeying into ERP or finance systems. E-procurement opportunity: Guided forms, catalog buying, auto-save, field validation, budget integration, and status dashboards reduce friction. Blockchain opportunity: A tamper-evident request timestamp discourages retroactive

manipulation and supports audit of who requested what and when.

3) *BP03: Purchase Request Approval: Objective.* Authorize spending before a sourcing event or purchase order is executed. *Trigger.* Submitted requisition requires approval *Actors.* The primary actor is Approver/Manager; supporting actors are Requesting Department, Procurement Officer, Finance Officer. *Preconditions.* Requisition is submitted; approval matrix, delegation rules, and budget checks are configured. *Inputs and outputs.* Inputs include Submitted requisition, approval matrix, budget availability, policy exceptions, supporting documents.. Outputs include Approval record, approved/rejected status, audit entry, next workflow task. *Normal flow.* 1) Backend routes the request to the correct approver by cost center, category, and value. 2) Approver reviews amount, justification,

TABLE VII
USE CASE SPECIFICATION SUMMARY (PART 2 OF 5)

Use Case ID	Use Case Name	Actor	Goal	Preconditions	Main Flow	Alternative Flow	Exception Flow	Postconditions
UC04	Request for Quotation Preparation and Publication	Procurement Officer	Translate an approved requirement into a controlled sourcing event with clear specifications and deadlines.	Requisition is approved; sourcing method and procurement threshold are known; supplier list or market search is available.	Procurement officer creates RFQ from the approved requisition.; System applies template, deadline, eligibility, and evaluation criteria.; Backend stores RFQ package and sends invitations through supplier portal.	For low-value catalog items, competitive RFQ may be bypassed under pre-approved catalog or framework agreement rules.	If specifications are ambiguous or approval has expired, the RFQ is returned for correction or re-approval.	RFQ record, published tender package, invited supplier notifications, RfqPublished event.
UC05	Supplier Quotation Submission	Vendor/Supplier	Allow qualified suppliers to submit commercial and technical offers in a controlled and auditable manner.	Supplier is approved or prequalified; RFQ is open; bid submission window and confidentiality rules are active.	Supplier opens RFQ in portal and prepares quotation.; Frontend validates mandatory bid fields and attachment constraints.; Backend stores quotation off-chain and prevents late modification after submission deadline.	Supplier may withdraw and resubmit before the deadline if RFQ rules permit versioning.	Late bids, unauthorized suppliers, or malformed submissions are rejected and logged.	Quotation record, encrypted/off-chain attachments, QuotationSubmitted event, bid receipt confirmation.
UC06	Quotation Evaluation and Supplier Award	Procurement Officer	Evaluate supplier offers fairly and select the best supplier according to predefined criteria.	RFQ is closed; quotations are opened according to policy; evaluation criteria are approved.	System opens eligible quotations after deadline.; Evaluation committee scores technical and commercial criteria.; Backend calculates ranking, flags anomalies, and prepares award recommendation.	If negotiations or clarifications are permitted, revised best-and-final offers are tracked as controlled versions.	Conflict of interest, abnormal price, missing compliance evidence, or supplier protest creates an exception case.	Evaluation scorecard, award decision, award approval task, AwardRecorded event.

TABLE VIII
USE CASE SPECIFICATION SUMMARY (PART 3 OF 5)

Use Case ID	Use Case Name	Actor	Goal	Preconditions	Main Flow	Alternative Flow	Exception Flow	Postconditions
UC07	Purchase Order Generation and Supplier Acceptance	Procurement Officer	Create a formal buying commitment specifying supplier, items, quantities, prices, delivery terms, and payment terms.	Supplier is active; requisition or award is approved; budget and contract terms are available.	Backend generates PO from approved source document.; Procurement officer reviews commercial and delivery terms.; System sends PO to supplier portal and stores ERP/finance reference.	For blanket or framework agreements, PO may be a release order against an existing contract.	If supplier rejects terms or price differs from award, the PO enters amendment or exception handling.	PurchaseOrder record, supplier acknowledgement, POIssued event, ERP posting.
UC08	Contract Creation and Approval	Procurement Officer	Formalize negotiated terms, responsibilities, prices, service levels, compliance clauses, and renewal/termination provisions.	Supplier award is approved; contract template and required clauses are available; legal review rules are defined.	Procurement generates contract draft from template and award data.; Legal and business approvers review clauses and risk terms.; Supplier reviews and signs through approved channel.	Standard low-risk terms may use pre-approved templates with accelerated approval.	Clause disagreement, missing signatory authority, or compliance issue sends the contract to negotiation or escalation.	Contract record, approved contract version, obligation register, ContractApproved event.
UC09	Goods or Service Delivery Confirmation	Receiving Officer	Verify that delivered goods or services conform to the PO/contract before invoice matching and payment.	PO is issued and acknowledged; receiving location or service owner is defined.	Supplier delivers goods or service evidence.; Receiver records delivery against PO lines and inspects quantity/quality.; Backend updates receiving status and available invoice matching data.	Partial delivery is allowed if PO terms permit split shipments or milestones.	Damaged goods, over-delivery, missing documents, or rejected service creates a discrepancy case.	DeliveryReceipt, accepted quantity, rejection reason, GoodsReceived event.

budget, and policy warnings. 3) Approver approves, rejects, or requests clarification. 4) Database records the decision; chaincode records decision digest and actor identity. 5) System advances requisition to RFQ or PO generation. *Alternative and exception flows.* Alternative: If the approver delegates authority, the delegated approver acts under an auditable delegation rule. Exception: If approval exceeds defined authority or segregation-of-duties rules fail, the workflow is blocked and escalated. *Business rules.* Requester cannot approve own request; high-value or restricted-category requests require multi-level approval. *Inefficiency and improvement.* Current pain point: Manual approvals are slow, sequential, opaque, and vulnerable to lost email approvals or informal authorization. E-

procurement opportunity: Approval queues, SLA timers, mobile approval, automated escalation, and policy-based routing shorten cycle time. Blockchain opportunity: Fabric captures the approval decision as a non-repudiable event linked to a verifiable organizational identity.

4) *BP04: Request for Quotation Preparation and Publication: Objective.* Translate an approved requirement into a controlled sourcing event with clear specifications and deadlines. *Trigger.* Approved requisition requires competitive quotation or tendering *Actors.* The primary actor is Procurement Officer; supporting actors are Requesting Department, Supplier, Approver. *Preconditions.* Requisition is approved; sourcing method and procurement threshold are known; sup-

TABLE IX
USE CASE SPECIFICATION SUMMARY (PART 4 OF 5)

Use Case ID	Use Case Name	Actor	Goal	Preconditions	Main Flow	Alternative Flow	Exception Flow	Postconditions
UC10	Invoice Submission	Vendor/Supplier	Capture supplier billing data in a structured form that can be matched and approved efficiently.	Supplier is active; PO or contract exists; delivery or service acceptance rules are satisfied or pending.	Supplier creates invoice from PO/receipt in portal or uploads e-invoice.; Frontend checks mandatory fields and PO reference.; Backend validates supplier, tax format, duplicate invoice number, and line totals.	A non-PO invoice may be routed through a stricter approval path if policy permits non-PO spend.	Duplicate invoice number, inactive supplier, mismatched bank data, or invalid tax data blocks submission.	Invoice record, validation result, InvoiceSubmitted event, AP work item.
UC11	Invoice Verification and Three-Way Matching	Finance Officer	Confirm that invoice, PO, and receipt/service acceptance agree before payment approval.	Invoice exists; PO/contract and goods receipt or service acceptance are available.	Backend retrieves invoice, PO, contract, and receipt data.; System compares supplier, items, quantities, prices, tax, and tolerances.; Clean matches are auto-approved for payment workflow.	Two-way matching may be used for service or low-risk non-stock purchases if policy allows.	Quantity variance, price variance, missing receipt, tax error, or duplicate invoice creates an exception case.	Match result, approved exception case, InvoiceMatched event.
UC12	Payment Approval and Payment Instruction	Finance Officer	Authorize payment to supplier after invoice verification while preserving segregation of duties and treasury controls.	Invoice is approved for payment; supplier bank data is validated; payment terms and cash schedule are available.	Finance reviews matched invoice and payment due date.; Backend checks segregation of duties, duplicate payment, bank data, and approval threshold.; Approver authorizes payment.	Early-payment discount workflow may prioritize payment before standard due date.	Bank validation failure, duplicate payment risk, cash hold, or sanctions alert prevents release.	PaymentApproval, payment instruction, status update, PaymentApproved or PaymentSettled event.

TABLE X
USE CASE SPECIFICATION SUMMARY (PART 5 OF 5)

Use Case ID	Use Case Name	Actor	Goal	Preconditions	Main Flow	Alternative Flow	Exception Flow	Postconditions
UC13	Procurement Audit and Compliance Review	Auditor/Compliance Officer	Assess whether procurement transactions followed policy, contract, approval, and regulatory requirements.	Process data, logs, approvals, and blockchain records exist for the selected period or transaction set.	Auditor selects a transaction or sample population.; System assembles end-to-end evidence from database and ledger.; Auditor reviews policy compliance, SoD, timestamps, and exceptions.	Continuous monitoring may automatically flag transactions for review based on risk rules.	Missing off-chain evidence, conflicting status, or suspected tampering triggers investigation.	AuditRecord, findings, control exceptions, remediation plan, AuditFindingRecorded event.
UC14	Dispute or Exception Handling	Procurement Officer	Resolve deviations from the normal procurement process without losing accountability or evidence.	An exception or dispute has been raised from RFQ, award, PO, delivery, invoice, payment, or audit process.	System creates exception case linked to source transaction.; Responsible parties review evidence and communicate through controlled messages.; Resolution owner accepts, rejects, corrects, or escalates the case.	Low-risk tolerance exceptions may be auto-approved if rules permit and all evidence is present.	Unresolved disputes, suspected fraud, or legal risk are escalated to compliance/legal and may suspend payment or supplier status.	ExceptionCase or DisputeCase, resolution decision, corrective action, ExceptionResolved event.
UC15	Supplier Performance and Contract Closeout	Procurement Officer	Use transaction history to measure supplier performance, close contracts formally, and improve future sourcing decisions.	PO/contract has delivery, invoice, or milestone data; KPIs and closeout criteria are defined.	System aggregates delivery, invoice, quality, and dispute metrics.; Procurement reviews supplier scorecard with business owner.; Supplier may acknowledge feedback or submit improvement plan.	High-performing suppliers may be flagged as preferred for future sourcing; poor performers may enter corrective action.	Unresolved claims, open invoices, or nonconformance prevents closeout until resolved.	SupplierPerformance scorecard, contract closeout, renewal recommendation, SupplierPerformanceRecorded event.

plier list or market search is available. *Inputs and outputs.* Inputs include Approved requisition, technical specification, commercial terms, evaluation criteria, bid deadline, invited supplier list.. Outputs include RFQ record, published tender package, invited supplier notifications, RfqPublished event. *Normal flow.* 1) Procurement officer creates RFQ from the approved requisition. 2) System applies template, deadline, eligibility, and evaluation criteria. 3) Backend stores RFQ package and sends invitations through supplier portal. 4) Chaincode records RFQ hash, issue timestamp, and supplier access rules. 5) Suppliers receive controlled access to submit quotations. *Alternative and exception flows.* Alternative: For low-value catalog items, competitive RFQ may be bypassed under pre-

approved catalog or framework agreement rules. Exception: If specifications are ambiguous or approval has expired, the RFQ is returned for correction or re-approval. *Business rules.* All suppliers must receive the same material RFQ information; changes after publication require addendum tracking. *Inefficiency and improvement.* Current pain point: Manual RFQ creation duplicates requisition data, creates inconsistent evaluation criteria, and makes bid-window disputes difficult to resolve. E-procurement opportunity: RFQ templates, auto-invitation, controlled clarifications, electronic deadlines, and supplier communication logs standardize sourcing. Blockchain opportunity: Publication hashes and addendum history provide tamper-evident proof of what was issued and when.

5) *BP05: Supplier Quotation Submission: Objective.* Allow qualified suppliers to submit commercial and technical offers in a controlled and auditable manner. *Trigger.* Supplier receives RFQ invitation or public tender access *Actors.* The primary actor is Vendor/Supplier; supporting actors are Procurement Officer, System Administrator. *Preconditions.* Supplier is approved or prequalified; RFQ is open; bid submission window and confidentiality rules are active. *Inputs and outputs.* Inputs include Price, delivery date, terms, technical response, attachments, certifications, alternative offer information.. Outputs include Quotation record, encrypted/off-chain attachments, QuotationSubmitted event, bid receipt confirmation. *Normal flow.* 1) Supplier opens RFQ in portal and prepares quotation. 2) Frontend validates mandatory bid fields and attachment constraints. 3) Backend stores quotation off-chain and prevents late modification after submission deadline. 4) Chaincode records bid hash, supplier identity, RFQ link, and submission timestamp. 5) System issues bid receipt without revealing competing quotations. *Alternative and exception flows.* Alternative: Supplier may withdraw and resubmit before the deadline if RFQ rules permit versioning. Exception: Late bids, unauthorized suppliers, or malformed submissions are rejected and logged. *Business rules.* Bid content remains confidential until the opening/evaluation stage; submitted bids are locked after the closing time. *Inefficiency and improvement.* Current pain point: Email submissions are vulnerable to timestamp disputes, accidental disclosure, version confusion, and missing attachments. E-procurement opportunity: Supplier portal, deadline control, mandatory field checks, and secure receipt confirmation improve fairness and traceability. Blockchain opportunity: Fabric records a tamper-evident bid commitment without storing sensitive commercial details on-chain.

6) *BP06: Quotation Evaluation and Supplier Award: Objective.* Evaluate supplier offers fairly and select the best supplier according to predefined criteria. *Trigger.* RFQ closes and quotations are available for evaluation *Actors.* The primary actor is Procurement Officer; supporting actors are Evaluation Committee, Requesting Department, Approver, Auditor. *Preconditions.* RFQ is closed; quotations are opened according to policy; evaluation criteria are approved. *Inputs and outputs.* Inputs include Quotations, scoring criteria, supplier qualification, delivery terms, total cost, compliance checks.. Outputs include Evaluation scorecard, award decision, award approval task, AwardRecorded event. *Normal flow.* 1) System opens eligible quotations after deadline. 2) Evaluation committee scores technical and commercial criteria. 3) Backend calculates ranking, flags anomalies, and prepares award recommendation. 4) Approver confirms award decision. 5) Chaincode records scorecard hash and award decision event. *Alternative and exception flows.* Alternative: If negotiations or clarifications are permitted, revised best-and-final offers are tracked as controlled versions. Exception: Conflict of interest, abnormal price, missing compliance evidence, or supplier protest creates an exception case. *Business rules.* Evaluation must use criteria published before bid submission; evaluators cannot modify submitted quotation contents. *Inefficiency and improvement.*

Current pain point: Spreadsheet evaluations can be subjective, hard to audit, and disconnected from RFQ rules and supplier records. E-procurement opportunity: Scoring templates, automatic comparison, committee workflow, and conflict declarations improve transparency. Blockchain opportunity: Award decision and evaluation hashes create immutable evidence for supplier protests and audits.

7) *BP07: Purchase Order Generation and Supplier Acceptance: Objective.* Create a formal buying commitment specifying supplier, items, quantities, prices, delivery terms, and payment terms. *Trigger.* Approved award, catalog requisition, or contract release *Actors.* The primary actor is Procurement Officer; supporting actors are Supplier, Finance Officer, Requesting Department. *Preconditions.* Supplier is active; requisition or award is approved; budget and contract terms are available. *Inputs and outputs.* Inputs include Approved requisition, quotation/contract, item lines, quantities, prices, tax codes, delivery address, payment terms.. Outputs include PurchaseOrder record, supplier acknowledgement, POIssued event, ERP posting. *Normal flow.* 1) Backend generates PO from approved source document. 2) Procurement officer reviews commercial and delivery terms. 3) System sends PO to supplier portal and stores ERP/finance reference. 4) Supplier accepts, rejects, or requests change. 5) Chaincode records PO issue and acknowledgement events. *Alternative and exception flows.* Alternative: For blanket or framework agreements, PO may be a release order against an existing contract. Exception: If supplier rejects terms or price differs from award, the PO enters amendment or exception handling. *Business rules.* PO cannot exceed approved budget or contract price unless an authorized amendment exists. *Inefficiency and improvement.* Current pain point: Manual PO creation causes rekeying errors, slow supplier communication, and mismatches with invoices. E-procurement opportunity: Auto-create PO, electronic acknowledgement, integration with ERP, and supplier portal reduce order cycle time. Blockchain opportunity: PO lifecycle events become tamper-evident commitments shared among buyer, supplier, and auditors.

8) *BP08: Contract Creation and Approval: Objective.* Formalize negotiated terms, responsibilities, prices, service levels, compliance clauses, and renewal/termination provisions. *Trigger.* Award requires formal contract or framework agreement *Actors.* The primary actor is Procurement Officer; supporting actors are Legal Officer, Supplier, Approver, Compliance Officer. *Preconditions.* Supplier award is approved; contract template and required clauses are available; legal review rules are defined. *Inputs and outputs.* Inputs include Award decision, negotiated terms, supplier details, price schedule, SLAs, compliance clauses, risk provisions.. Outputs include Contract record, approved contract version, obligation register, ContractApproved event. *Normal flow.* 1) Procurement generates contract draft from template and award data. 2) Legal and business approvers review clauses and risk terms. 3) Supplier reviews and signs through approved channel. 4) Backend stores full contract off-chain and extracts obligations. 5) Chaincode records version hash, approval identities,

active state, and amendment references. *Alternative and exception flows.* Alternative: Standard low-risk terms may use pre-approved templates with accelerated approval. Exception: Clause disagreement, missing signatory authority, or compliance issue sends the contract to negotiation or escalation. *Business rules.* Only approved contract versions may be linked to POs and invoices; amendments require new version hashes. *Inefficiency and improvement.* Current pain point: Contracts are often stored outside buying workflows, causing price leakage, missed renewals, and weak obligation tracking. E-procurement opportunity: CLM integration, approval routing, obligation extraction, and renewal notifications strengthen contract control. Blockchain opportunity: Contract hash and amendment chronology prove which version was active at each procurement event.

9) *BP09: Goods or Service Delivery Confirmation: Objective.* Verify that delivered goods or services conform to the PO/contract before invoice matching and payment. *Trigger.* Supplier delivers goods or completes service milestone *Actors.* The primary actor is Receiving Officer; supporting actors are Supplier, Requesting Department, Procurement Officer. *Preconditions.* PO is issued and acknowledged; receiving location or service owner is defined. *Inputs and outputs.* Inputs include Delivery note, PO number, delivered quantities, inspection results, photos, service completion evidence.. Outputs include DeliveryReceipt, accepted quantity, rejection reason, GoodsReceived event. *Normal flow.* 1) Supplier delivers goods or service evidence. 2) Receiver records delivery against PO lines and inspects quantity/quality. 3) Backend updates receiving status and available invoice matching data. 4) Discrepancies are sent to procurement/supplier for correction. 5) Chaincode records accepted receipt hash and delivery status event. *Alternative and exception flows.* Alternative: Partial delivery is allowed if PO terms permit split shipments or milestones. Exception: Damaged goods, over-delivery, missing documents, or rejected service creates a discrepancy case. *Business rules.* Invoice cannot be fully approved without accepted delivery or authorized service acceptance. *Inefficiency and improvement.* Current pain point: Paper receiving notes are lost or delayed, creating invoice blocks and payment disputes. E-procurement opportunity: Mobile receiving, PO line matching, photo evidence, and real-time status updates reduce invoice exceptions. Blockchain opportunity: Delivery confirmation becomes a shared tamper-evident milestone for buyer-supplier dispute resolution.

10) *BP10: Invoice Submission: Objective.* Capture supplier billing data in a structured form that can be matched and approved efficiently. *Trigger.* Supplier has delivered goods/services and seeks payment *Actors.* The primary actor is Vendor/-Supplier; supporting actors are Finance Officer, Procurement Officer. *Preconditions.* Supplier is active; PO or contract exists; delivery or service acceptance rules are satisfied or pending. *Inputs and outputs.* Inputs include Invoice number, supplier tax data, PO/contract reference, line amounts, taxes, bank/payment details, attachments.. Outputs include Invoice record, validation result, InvoiceSubmitted event, AP work

item. *Normal flow.* 1) Supplier creates invoice from PO/receipt in portal or uploads e-invoice. 2) Frontend checks mandatory fields and PO reference. 3) Backend validates supplier, tax format, duplicate invoice number, and line totals. 4) Database stores invoice data and attachments off-chain. 5) Chaincode records invoice hash and submission event. *Alternative and exception flows.* Alternative: A non-PO invoice may be routed through a stricter approval path if policy permits non-PO spend. Exception: Duplicate invoice number, inactive supplier, mismatched bank data, or invalid tax data blocks submission. *Business rules.* Invoice cannot be paid unless matched or explicitly approved as an exception. *Inefficiency and improvement.* Current pain point: Email invoices cause lost invoices, duplicate payments, weak supplier visibility, and slow AP queues. E-procurement opportunity: PO-flip e-invoicing, duplicate checks, structured tax validation, and self-service status reduce AP workload. Blockchain opportunity: Invoice submission timestamp and hash support non-repudiation without exposing confidential invoice images on-chain.

11) *BP11: Invoice Verification and Three-Way Matching: Objective.* Confirm that invoice, PO, and receipt/service acceptance agree before payment approval. *Trigger.* Invoice is submitted and ready for validation *Actors.* The primary actor is Finance Officer; supporting actors are Procurement Officer, Receiving Officer, Supplier. *Preconditions.* Invoice exists; PO/contract and goods receipt or service acceptance are available. *Inputs and outputs.* Inputs include Invoice, PO, goods receipt/service acceptance, tax data, tolerance thresholds, exception history.. Outputs include Match result, approved invoice status or exception case, InvoiceMatched event. *Normal flow.* 1) Backend retrieves invoice, PO, contract, and receipt data. 2) System compares supplier, items, quantities, prices, tax, and tolerances. 3) Clean matches are auto-approved for payment workflow. 4) Exceptions are routed to responsible role for resolution. 5) Chaincode records match result hash and decision event. *Alternative and exception flows.* Alternative: Two-way matching may be used for service or low-risk non-stock purchases if policy allows. Exception: Quantity variance, price variance, missing receipt, tax error, or duplicate invoice creates an exception case. *Business rules.* Tolerance thresholds determine auto-match eligibility; override requires documented authority. *Inefficiency and improvement.* Current pain point: Manual matching is labor-intensive and creates delayed payments, supplier disputes, and duplicate payment risk. E-procurement opportunity: Automated matching, tolerance rules, and exception routing reduce AP cost and cycle time. Blockchain opportunity: A recorded match decision and exception audit trail improve accountability for overrides and disputed payments.

12) *BP12: Payment Approval and Payment Instruction: Objective.* Authorize payment to supplier after invoice verification while preserving segregation of duties and treasury controls. *Trigger.* Invoice is matched or exception approval is complete *Actors.* The primary actor is Finance Officer; supporting actors are Approver/Manager, Supplier, Treasury/Bank Interface. *Preconditions.* Invoice is approved for payment; sup-

plier bank data is validated; payment terms and cash schedule are available. *Inputs and outputs.* Inputs include Approved invoice, payment terms, bank account, payment amount, cash forecast, approval matrix.. Outputs include PaymentApproval, payment instruction, status update, PaymentApproved or PaymentSettled event. *Normal flow.* 1) Finance reviews matched invoice and payment due date. 2) Backend checks segregation of duties, duplicate payment, bank data, and approval threshold. 3) Approver authorizes payment. 4) Payment instruction is sent to ERP/treasury/bank interface. 5) Chaincode records payment approval hash and settlement status when available. *Alternative and exception flows.* Alternative: Early-payment discount workflow may prioritize payment before standard due date. Exception: Bank validation failure, duplicate payment risk, cash hold, or sanctions alert prevents release. *Business rules.* Payment approver cannot be the same user who created supplier bank data or approved the requisition if policy forbids it. *Inefficiency and improvement.* Current pain point: Manual payment approval can miss discounts, create late fees, and weaken segregation-of-duties evidence. E-procurement opportunity: Automated payment worklists, discount alerts, bank integration, and SoD checks improve control and supplier trust. Blockchain opportunity: Payment approval events support an immutable final audit trail, while actual banking data remains off-chain.

13) *BP13: Procurement Audit and Compliance Review:* *Objective.* Assess whether procurement transactions followed policy, contract, approval, and regulatory requirements. *Trigger.* Scheduled audit, risk alert, dispute, management review, or regulatory request *Actors.* The primary actor is Auditor/Compliance Officer; supporting actors are Procurement Officer, Finance Officer, System Administrator. *Preconditions.* Process data, logs, approvals, and blockchain records exist for the selected period or transaction set. *Inputs and outputs.* Inputs include Requisition, approval, RFQ, quotation, PO, contract, receipt, invoice, payment, user logs, ledger events.. Outputs include AuditRecord, findings, control exceptions, remediation plan, AuditFindingRecorded event. *Normal flow.* 1) Auditor selects a transaction or sample population. 2) System assembles end-to-end evidence from database and ledger. 3) Auditor reviews policy compliance, SoD, timestamps, and exceptions. 4) Findings are recorded and assigned for remediation. 5) Chaincode records audit finding hash and closure event. *Alternative and exception flows.* Alternative: Continuous monitoring may automatically flag transactions for review based on risk rules. Exception: Missing off-chain evidence, conflicting status, or suspected tampering triggers investigation. *Business rules.* Auditors receive read-only access; audit finding edits require versioned correction rather than deletion. *Inefficiency and improvement.* Current pain point: Manual audit evidence collection is slow, incomplete, and dependent on scattered email/files. E-procurement opportunity: Audit dashboards, electronic evidence packs, exception reports, and workflow closure improve audit efficiency. Blockchain opportunity: Fabric ledger entries provide a tamper-evident chronological backbone for audit and compliance evidence.

14) *BP14: Dispute or Exception Handling: Objective.* Resolve deviations from the normal procurement process without losing accountability or evidence. *Trigger.* Mismatch, rejected delivery, supplier protest, invoice exception, contract dispute, or policy violation *Actors.* The primary actor is Procurement Officer; supporting actors are Supplier, Finance Officer, Receiving Officer, Auditor, Approver. *Preconditions.* An exception or dispute has been raised from RFQ, award, PO, delivery, invoice, payment, or audit process. *Inputs and outputs.* Inputs include Exception reason, affected transaction, supporting evidence, messages, user comments, resolution authority.. Outputs include ExceptionCase or DisputeCase, resolution decision, corrective action, ExceptionResolved event. *Normal flow.* 1) System creates exception case linked to source transaction. 2) Responsible parties review evidence and communicate through controlled messages. 3) Resolution owner accepts, rejects, corrects, or escalates the case. 4) Backend updates affected transaction status. 5) Chaincode records resolution hash, decision, and responsible actor. *Alternative and exception flows.* Alternative: Low-risk tolerance exceptions may be auto-approved if rules permit and all evidence is present. Exception: Unresolved disputes, suspected fraud, or legal risk are escalated to compliance/legal and may suspend payment or supplier status. *Business rules.* Corrections must preserve original records; no transaction history is deleted. *Inefficiency and improvement.* Current pain point: Email-based exception handling lacks ownership, SLA visibility, and reliable linkages to original documents. E-procurement opportunity: Case management, SLA alerts, structured reasons, and root-cause reports reduce unresolved exceptions. Blockchain opportunity: A shared evidence trail helps demonstrate when issues were raised, who responded, and how they were resolved.

15) *BP15: Supplier Performance and Contract Closeout: Objective.* Use transaction history to measure supplier performance, close contracts formally, and improve future sourcing decisions. *Trigger.* Delivery completion, contract milestone, periodic review, or contract expiry *Actors.* The primary actor is Procurement Officer; supporting actors are Requesting Department, Supplier, Compliance Officer, Auditor. *Preconditions.* PO/contract has delivery, invoice, or milestone data; KPIs and closeout criteria are defined. *Inputs and outputs.* Inputs include Delivery timeliness, quality results, invoice exceptions, dispute history, SLA performance, user feedback, audit findings.. Outputs include SupplierPerformance scorecard, contract closeout, renewal recommendation, SupplierPerformanceRecorded event. *Normal flow.* 1) System aggregates delivery, invoice, quality, and dispute metrics. 2) Procurement reviews supplier scorecard with business owner. 3) Supplier may acknowledge feedback or submit improvement plan. 4) Closeout decision is recorded for completed contracts. 5) Chaincode records scorecard digest and closeout decision. *Alternative and exception flows.* Alternative: High-performing suppliers may be flagged as preferred for future sourcing; poor performers may enter corrective action. Exception: Unresolved claims, open invoices, or nonconformance prevents closeout until resolved. *Business rules.* Performance data should be

based on recorded transaction evidence; supplier access to detailed internal notes may be restricted. *Inefficiency and improvement.* Current pain point: Supplier performance is often anecdotal and disconnected from actual P2P/S2C transaction records. E-procurement opportunity: Analytics-based scorecards and closeout workflow improve supplier relationship management and sourcing decisions. Blockchain opportunity: Ledger-backed milestones make scorecards harder to manipulate and more defensible in supplier reviews.

IX. UML BEHAVIORAL DIAGRAM PACKAGE

The diagram package contains two overall diagrams, one component architecture diagram, and three process-specific behavioral diagrams for each selected process: an activity diagram, a sequence diagram, and a state machine. The diagrams are supplied as PlantUML source to remain readable, editable, and version-control friendly. Table XI lists the diagram inventory. Appendix A contains the source listings.

A. Diagram Interpretation

The use case diagram defines what the system must do from each actor's perspective; it is not intended to show step order. Activity diagrams show step order, decision points, and exception routes. Sequence diagrams expose architectural responsibility by showing how a user action travels through frontend, backend, database, Fabric client SDK, chaincode, peers, ordering service, and ledger. State machines show the controlled lifecycle of procurement assets. These lifecycle models are critical for Fabric chaincode because chaincode must reject invalid transitions, such as paying an invoice before it is matched or issuing a PO to an unapproved supplier.

X. E-PROCUREMENT AND BLOCKCHAIN SYSTEM MAPPING

Table XII maps each process to candidate system components. The mapping is intentionally conceptual: implementation teams may combine or split services depending on scale, ERP constraints, cloud platform, and security architecture.

A. Off-Chain and On-Chain Data Boundary

The operational database should store mutable workflow data, search indexes, user interface metadata, large attachments, full contract documents, invoice images, tax files, bank files, and sensitive commercial details. The Fabric ledger should store minimal non-sensitive metadata, asset identifiers, lifecycle states, timestamps, submitting identity, decision identity, document hashes, and event references. This boundary limits confidentiality and scalability risks while preserving non-repudiation and traceability.

XI. HYPERLEDGER FABRIC ARCHITECTURE MAPPING

A procurement-oriented Fabric network can be organized around buyer, supplier, auditor, and possibly finance or logistics organizations. Each organization receives identities from a Fabric CA or trusted CA and is represented in MSP configuration. A channel may be created for the buyer-supplier procurement network. For highly sensitive bid or contract

data, private data collections should be considered so that only authorized organizations share private state while hashes remain on the channel ledger. Chaincode should implement procurement asset lifecycle rules and should not replace the operational database.

A. Participants and Identities

Buyer employees, supplier users, auditors, and system services should be authenticated through enterprise identity management and mapped to Fabric identities where ledger transactions are required. Fabric identities should carry organizational membership and role information sufficient for endorsement and access-control decisions. A practical design may allow the backend service to submit transactions on behalf of authenticated users while preserving user identity in signed transaction metadata and application audit records.

B. Peers, Ordering Service, Ledger, and Chaincode

Endorsing peers execute chaincode and return endorsement responses. The ordering service orders endorsed transactions without evaluating business semantics. Peers then validate endorsements and commit valid blocks to the channel ledger. Chaincode functions such as `RegisterVendor`, `CreateRequisition`, `SubmitQuotation`, `IssuePO`, `VerifyInvoiceMatch`, and `ApprovePayment` should validate current asset state, actor role, required references, and transition rules. Table XIII lists candidate assets.

C. Endorsement and Privacy Design

Endorsement policies should reflect business risk. For example, vendor approval may require buyer procurement and compliance endorsement; quotation submission may require supplier endorsement and buyer network validation; award decisions may require buyer procurement and audit-visible recording; payment approval may require finance endorsement. Private data collections are relevant for bank details, detailed quotation prices, commercially sensitive contract schedules, and supplier-specific data. For many processes, storing only a cryptographic hash and lifecycle event on-chain is sufficient.

XII. MANUAL VERSUS DIGITIZED AND BLOCKCHAIN-SUPPORTED PROCESS

In the manual or semi-manual process, evidence is distributed across email, documents, spreadsheets, paper signatures, and disconnected systems. This creates rekeying, weak status visibility, delayed approvals, inconsistent supplier data, and expensive audit reconstruction. In the proposed digitized process, frontend screens capture structured data once; backend services validate workflow rules and orchestrate integration; databases retain operational records and documents; and Fabric records selected lifecycle events as a tamper-evident audit backbone. The improvement does not arise from blockchain alone. It arises from the combination of process standardization, guided user interfaces, workflow automation, integration, data governance, and ledger-backed evidence.

TABLE XI
PLANTUML DIAGRAM INVENTORY

ID	Process	Activity diagram listing	Sequence diagram listing	State-machine listing
BP01	Vendor Registration and Onboarding	BP01_activity	BP01_sequence	BP01_state
BP02	Purchase Requisition Creation	BP02_activity	BP02_sequence	BP02_state
BP03	Purchase Request Approval	BP03_activity	BP03_sequence	BP03_state
BP04	Request for Quotation Preparation and Publication	BP04_activity	BP04_sequence	BP04_state
BP05	Supplier Quotation Submission	BP05_activity	BP05_sequence	BP05_state
BP06	Quotation Evaluation and Supplier Award	BP06_activity	BP06_sequence	BP06_state
BP07	Purchase Order Generation and Supplier Acceptance	BP07_activity	BP07_sequence	BP07_state
BP08	Contract Creation and Approval	BP08_activity	BP08_sequence	BP08_state
BP09	Goods or Service Delivery Confirmation	BP09_activity	BP09_sequence	BP09_state
BP10	Invoice Submission	BP10_activity	BP10_sequence	BP10_state
BP11	Invoice Verification and Three-Way Matching	BP11_activity	BP11_sequence	BP11_state
BP12	Payment Approval and Payment Instruction	BP12_activity	BP12_sequence	BP12_state
BP13	Procurement Audit and Compliance Review	BP13_activity	BP13_sequence	BP13_state
BP14	Dispute or Exception Handling	BP14_activity	BP14_sequence	BP14_state
BP15	Supplier Performance and Contract Closeout	BP15_activity	BP15_sequence	BP15_state

XIII. DISCUSSION

The main design implication is that procurement should be modeled as a set of connected asset lifecycles rather than isolated forms. VendorProfile enables RFQ and PO; PurchaseRequisition leads to approval, RFQ, or PO; Quotation leads to award; PurchaseOrder leads to delivery and invoice; Invoice leads to matching and payment; AuditRecord and DisputeCase can reference every prior asset. This traceability is valuable for engineers because it identifies where APIs, database keys, message events, and chaincode transitions must align.

Blockchain should be used selectively. It is useful where multiple parties need shared evidence of event chronology, where a supplier may dispute submission time or PO acknowledgement, where auditors need tamper-evident approval trails, or where contract version hashes must be linked to subsequent POs and invoices. It is less useful for high-volume mutable data, large documents, user interface drafts, search-heavy analytics, and confidential commercial details that do not need cross-party replication. Overusing blockchain can increase cost, complexity, privacy exposure, and operational latency.

The e-procurement architecture also depends on non-blockchain capabilities: master-data governance, catalog quality, ERP integration, role engineering, workflow configuration, supplier adoption, change management, and cybersecurity. A poorly configured digital workflow can reproduce manual inefficiencies at higher technical cost. Therefore, the process models in this paper should be treated as requirements artifacts that must be validated with stakeholders before implementation.

XIV. RECOMMENDATIONS

Implementation teams should begin with a process baseline before selecting software. Recommended steps are: (i) confirm

actor roles and approval thresholds; (ii) standardize supplier, item, contract, PO, receipt, invoice, and payment identifiers; (iii) define which events are material enough for ledger recording; (iv) store full documents and sensitive values off-chain; (v) design chaincode around asset lifecycle state transitions; (vi) configure endorsement policies by risk; (vii) introduce supplier self-service gradually; and (viii) use audit dashboards to prove process value.

For initial engineering work, the most valuable minimum viable product is likely to include vendor onboarding, requisition approval, PO issuance, delivery confirmation, invoice matching, and audit trail views. RFQ, contract management, payment integration, and supplier performance analytics can be layered once the data model and identity architecture are stable.

XV. LIMITATIONS

The study is generic and does not use transaction logs from a real organization. It does not estimate implementation cost, performance throughput, storage overhead, legal enforceability of digital signatures, local tax compliance, or integration complexity with a specific ERP. The UML diagrams are conceptual and should be refined into executable workflow specifications, API contracts, database schemas, and chaincode test cases. The references support general concepts and design reasoning; organization-specific benchmarks require primary data.

XVI. CONCLUSION

This paper documents procurement processes as behavioral models that bridge business analysis and system architecture. It identifies actors, terminology, process flows, business rules, inefficiencies, and improvement opportunities across supplier onboarding, requisitioning, approvals, sourcing, quotation, PO, contract, delivery, invoice, payment, audit, exception, and

TABLE XII
SYSTEM MAPPING TABLE

Business Step	Frontend Interface	Backend Service/API	Database Entity	Blockchain Asset	Chaincode Function	Ledger Event
BP01 Vendor Registration and Onboarding	Supplier Portal: Registration Screen	VendorService API	Vendor, VendorDocument, RiskCheck	VendorProfile	RegisterVendor, ApproveVendor, SuspendVendor	VendorRegistered, VendorApproved, VendorSuspended
BP02 Purchase Requisition Creation	Requester Web/Mobile Requisition Form	RequisitionService API	Requisition, RequisitionLine, BudgetReservation	PurchaseRequisition	CreateRequisition, AmendRequisition	RequisitionCreated, RequisitionAmended
BP03 Purchase Request Approval	Manager Approval Worklist	ApprovalWorkflowService API	ApprovalTask, DelegationRule, PolicyCheck	ApprovalDecision	RecordApproval, RejectRequisition, EscalateApproval	ApprovalRecorded, ApprovalRejected, ApprovalEscalated
BP04 Request for Quotation Preparation and Publication	Procurement RFQ Workspace	SourcingService API	RFQ, RFQLine, SupplierInvitation	RFQ	CreateRFQ, PublishRFQ, AmendRFQ	RFQCreated, RFQPublished, RFQAmended
BP05 Supplier Quotation Submission	Supplier Quotation Portal	QuotationService API	Quotation, QuotationAttachment, BidReceipt	Quotation	SubmitQuotation, WithdrawQuotation	QuotationSubmitted, QuotationWithdrawn
BP06 Quotation Evaluation and Supplier Award	Evaluation Committee Dashboard	EvaluationService API	EvaluationScore, AwardRecommendation, COIDeclaration	AwardDecision	RecordEvaluation, AwardSupplier	EvaluationRecorded, SupplierAwarded
BP07 Purchase Order Generation and Supplier Acceptance	PO Management Workspace	PurchaseOrderService API	PurchaseOrder, POLine, ERPPosting	PurchaseOrder	IssuePO, AcknowledgePO, AmendPO	POIssued, POAcknowledged, POAmended
BP08 Contract Creation and Approval	Contract Lifecycle Workspace	ContractService API	Contract, ContractVersion, Obligation	Contract	CreateContract, ApproveContract, AmendContract, ExpireContract	ContractCreated, ContractApproved, ContractAmended
BP09 Goods or Service Delivery Confirmation	Receiving Mobile/Web App	ReceivingService API	GoodsReceipt, InspectionRecord, DeliveryAttachment	DeliveryReceipt	RecordDelivery, AcceptDelivery, RejectDelivery	DeliveryRecorded, DeliveryAccepted, DeliveryRejected
BP10 Invoice Submission	Supplier Invoice Portal	InvoiceService API	Invoice, InvoiceLine, InvoiceAttachment	Invoice	SubmitInvoice, CancelInvoice	InvoiceSubmitted, InvoiceCancelled
BP11 Invoice Verification and Three-Way Matching	AP Invoice Verification Workbench	MatchingService API	MatchResult, ExceptionCase, ToleranceRule	InvoiceMatch	VerifyInvoiceMatch, RecordExceptionResolution	InvoiceMatched, InvoiceExceptionRaised, InvoiceExceptionResolved
BP12 Payment Approval and Payment Instruction	Finance Payment Approval Screen	PaymentService API	PaymentApproval, PaymentInstruction, BankValidation	PaymentApproval	ApprovePayment, RecordPaymentSettlement	PaymentApproved, PaymentInstructionSent, PaymentSettled
BP13 Procurement Audit and Compliance Review	Audit and Compliance Dashboard	AuditService API	AuditRecord, AuditFinding, ControlTest	AuditRecord	CreateAuditRecord, RecordFinding, CloseFinding	AuditRecordCreated, AuditFindingRecorded, AuditFindingClosed
BP14 Dispute or Exception Handling	Exception/Dispute Case Workspace	ExceptionService API	ExceptionCase, CaseMessage, CorrectiveAction	DisputeCase	OpenDispute, ResolveDispute, EscalateDispute	DisputeOpened, DisputeResolved, DisputeEscalated
BP15 Supplier Performance and Contract Closeout	Supplier Performance Dashboard	PerformanceService API	SupplierScorecard, KPIResult, ContractCloseout	SupplierPerformance	RecordPerformance, CloseContract	SupplierPerformanceRecorded, ContractClosed

supplier performance processes. It maps each process to e-procurement interfaces, backend services, database entities, Fabric assets, chaincode functions, and ledger events. The central conclusion is that e-procurement streamlines procurement through structured data capture, workflow automation, integration, analytics, and supplier self-service, while Hyperledger Fabric is best used as a permissioned audit and trust layer for material, multi-party procurement milestones.

TABLE XIII
HYPERLEDGER FABRIC PROCUREMENT ASSET LIFECYCLE MAPPING

Asset	Related Process	Ownership	Lifecycle States	Chaincode Functions	Events
VendorProfile	BP01 Vendor Registration and Onboarding	Vendor/Supplier initiates; buyer organization governs lifecycle	Draft, Submitted, UnderReview, Approved, Active, Suspended, Rejected, Archived	RegisterVendor, ApproveVendor, SuspendVendor	VendorRegistered, VendorApproved, VendorSuspended
PurchaseRequisition	BP02 Purchase Requisition Creation	Requesting Department initiates; buyer organization governs lifecycle	Draft, Submitted, UnderValidation, PendingApproval, Returned, Approved, Rejected, Cancelled	CreateRequisition, AmendRequisition	RequisitionCreated, RequisitionAmended
ApprovalDecision	BP03 Purchase Request Approval	Approver/Manager initiates; buyer organization governs lifecycle	Pending, InReview, ClarificationRequested, Approved, Rejected, Escalated, Expired	RecordApproval, RejectRequisition, EscalateApproval	ApprovalRecorded, ApprovalRejected, ApprovalEscalated
RFQ	BP04 Request for Quotation Preparation and Publication	Procurement Officer initiates; buyer organization governs lifecycle	Draft, InternalReview, Published, OpenForBids, Closed, Cancelled, Awarded	CreateRFQ, PublishRFQ, AmendRFQ	RFQCreated, RFQPublished, RFQAmended
Quotation	BP05 Supplier Quotation Submission	Vendor/Supplier initiates; buyer organization governs lifecycle	Draft, Submitted, Withdrawn, Opened, UnderEvaluation, Accepted, Rejected, Expired	SubmitQuotation, WithdrawQuotation	QuotationSubmitted, QuotationWithdrawn
AwardDecision	BP06 Quotation Evaluation and Supplier Award	Procurement Officer initiates; buyer organization governs lifecycle	NotStarted, Opened, Scoring, Clarification, Recommended, Approved, Rejected, Awarded, Protested	RecordEvaluation, AwardSupplier	EvaluationRecorded, SupplierAwarded
PurchaseOrder	BP07 Purchase Order Generation and Supplier Acceptance	Procurement Officer initiates; buyer organization governs lifecycle	Draft, Issued, Acknowledged, ChangeRequested, Amended, PartiallyDelivered, Delivered, Closed, Cancelled	IssuePO, AcknowledgePO, AmendPO	POIssued, POAcknowledged, POAmended
Contract	BP08 Contract Creation and Approval	Procurement Officer initiates; buyer organization governs lifecycle	Draft, LegalReview, SupplierReview, Approved, Active, Amended, Suspended, Expired, Terminated	CreateContract, ApproveContract, AmendContract, ExpireContract	ContractCreated, ContractApproved, ContractAmended
DeliveryReceipt	BP09 Goods or Service Delivery Confirmation	Receiving Officer initiates; buyer organization governs lifecycle	AwaitingDelivery, Delivered, UnderInspection, Accepted, PartiallyAccepted, Rejected, Closed	RecordDelivery, AcceptDelivery, RejectDelivery	DeliveryRecorded, DeliveryAccepted, DeliveryRejected
Invoice	BP10 Invoice Submission	Vendor/Supplier initiates; buyer organization governs lifecycle	Draft, Submitted, Validated, DuplicateSuspect, Matched, Exception, Approved, Paid, Cancelled	SubmitInvoice, CancelInvoice	InvoiceSubmitted, InvoiceCancelled
InvoiceMatch	BP11 Invoice Verification and Three-Way Matching	Finance Officer initiates; buyer organization governs lifecycle	ReadyForMatch, Matching, Matched, Exception, Clarification, Approved, Rejected	VerifyInvoiceMatch, RecordExceptionResolution	InvoiceMatched, InvoiceExceptionRaised, InvoiceExceptionResolved
PaymentApproval	BP12 Payment Approval and Payment Instruction	Finance Officer initiates; buyer organization governs lifecycle	PendingApproval, Approved, InstructionSent, Settled, Rejected, OnHold, Cancelled	ApprovePayment, RecordPaymentSettlement	PaymentApproved, PaymentInstructionSent, PaymentSettled
AuditRecord	BP13 Procurement Audit and Compliance Review	Auditor/Compliance Officer initiates; buyer organization governs lifecycle	Planned, EvidenceCollection, UnderReview, FindingRaised, Remediation, Closed, Escalated	CreateAuditRecord, RecordFinding, CloseFinding	AuditRecordCreated, AuditFindingRecorded, AuditFindingClosed
DisputeCase	BP14 Dispute or Exception Handling	Procurement Officer initiates; buyer organization governs lifecycle	Open, Assigned, UnderInvestigation, WaitingForSupplier, Resolved, Rejected, Escalated, Closed	OpenDispute, ResolveDispute, EscalateDispute	DisputeOpened, DisputeResolved, DisputeEscalated
SupplierPerformance	BP15 Supplier Performance and Contract Closeout	Procurement Officer initiates; buyer organization governs lifecycle	DataCollection, Scored, UnderReview, Acknowledged, ImprovementPlan, Preferred, Suspended, Closed	RecordPerformance, CloseContract	SupplierPerformanceRecorded, ContractClosed

TABLE XIV
PAIN POINT AND IMPROVEMENT TABLE

Current Pain Point	Business Impact	E-Procurement Improvement	Blockchain Improvement	Expected Benefit
Manual onboarding causes duplicate supplier records, inconsistent compliance checks, email-based document loss, and weak evidence for later audit.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Supplier portal, workflow routing, automatic completeness checks, risk APIs, and master-data governance reduce cycle time and rework.	Fabric provides non-repudiable onboarding events, status history, identity-based access control, and tamper-evident evidence hashes.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Email or spreadsheet requisitions are incomplete, hard to track, and often require rekeying into ERP or finance systems.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Guided forms, catalog buying, auto-save, field validation, budget integration, and status dashboards reduce friction.	A tamper-evident request timestamp discourages retroactive manipulation and supports audit of who requested what and when.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual approvals are slow, sequential, opaque, and vulnerable to lost email approvals or informal authorization.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Approval queues, SLA timers, mobile approval, automated escalation, and policy-based routing shorten cycle time.	Fabric captures the approval decision as a non-repudiable event linked to a verifiable organizational identity.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual RFQ creation duplicates requisition data, creates inconsistent evaluation criteria, and makes bid-window disputes difficult to resolve.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	RFQ templates, auto-invitation, controlled clarifications, electronic deadlines, and supplier communication logs standardize sourcing.	Publication hashes and addendum history provide tamper-evident proof of what was issued and when.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Email submissions are vulnerable to timestamp disputes, accidental disclosure, version confusion, and missing attachments.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Supplier portal, deadline control, mandatory field checks, and secure receipt confirmation improve fairness and traceability.	Fabric records a tamper-evident bid commitment without storing sensitive commercial details on-chain.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Spreadsheet evaluations can be subjective, hard to audit, and disconnected from RFQ rules and supplier records.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Scoring templates, automatic comparison, committee workflow, and conflict declarations improve transparency.	Award decision and evaluation hashes create immutable evidence for supplier protests and audits.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual PO creation causes rekeying errors, slow supplier communication, and mismatches with invoices.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Auto-create PO, electronic acknowledgement, integration with ERP, and supplier portal reduce order cycle time.	PO lifecycle events become tamper-evident commitments shared among buyer, supplier, and auditors.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Contracts are often stored outside buying workflows, causing price leakage, missed renewals, and weak obligation tracking.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	CLM integration, approval routing, obligation extraction, and renewal notifications strengthen contract control.	Contract hash and amendment chronology prove which version was active at each procurement event.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Paper receiving notes are lost or delayed, creating invoice blocks and payment disputes.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Mobile receiving, PO line matching, photo evidence, and real-time status updates reduce invoice exceptions.	Delivery confirmation becomes a shared tamper-evident milestone for buyer-supplier dispute resolution.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Email invoices cause lost invoices, duplicate payments, weak supplier visibility, and slow AP queues.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	PO-flip e-invoicing, duplicate checks, structured tax validation, and self-service status reduce AP workload.	Invoice submission timestamp and hash support non-repudiation without exposing confidential invoice images on-chain.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual matching is labor-intensive and creates delayed payments, supplier disputes, and duplicate payment risk.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Automated matching, tolerance rules, and exception routing reduce AP cost and cycle time.	A recorded match decision and exception audit trail improve accountability for overrides and disputed payments.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual payment approval can miss discounts, create late fees, and weaken segregation-of-duties evidence.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Automated payment worklists, discount alerts, bank integration, and SoD checks improve control and supplier trust.	Payment approval events support an immutable final audit trail, while actual banking data remains off-chain.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Manual audit evidence collection is slow, incomplete, and dependent on scattered email/files.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Audit dashboards, electronic evidence packs, exception reports, and workflow closure improve audit efficiency.	Fabric ledger entries provide a tamper-evident chronological backbone for audit and compliance evidence.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Email-based exception handling lacks ownership, SLA visibility, and reliable linkages to original documents.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Case management, SLA alerts, structured reasons, and root-cause reports reduce unresolved exceptions.	A shared evidence trail helps demonstrate when issues were raised, who responded, and how they were resolved.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.
Supplier performance is often anecdotal and disconnected from actual P2P/S2C transaction records.	Cycle-time delay, rework, weak control, audit difficulty, supplier friction, or cost leakage depending on the process.	Analytics-based scorecards and closeout workflow improve supplier relationship management and sourcing decisions.	Ledger-backed milestones make scorecards harder to manipulate and more defensible in supplier reviews.	Reduced manual handling, stronger controls, better traceability, and clearer accountability.

APPENDIX A PLANTUML SOURCE LISTINGS

The following listings provide editable PlantUML source for all diagrams in the diagram package. They can be rendered separately using PlantUML if graphical images are required.

Overall actor-system interaction context

```
1 @startuml
2 ' Diagram: Overall actor-system interaction context
3 ' Purpose: Show how business actors, application layers,
4   database, and Fabric network cooperate.
5 title Overall Actor-System Interaction Context
6 left to right direction
7 actor "Requesting\nDepartment" as Requester
8 actor "Approver/\nManager" as Approver
9 actor "Procurement\nOfficer" as Procurement
10 actor "Vendor/\nSupplier" as Supplier
11 actor "Finance/AP\nOfficer" as Finance
12 actor "Auditor/\nCompliance" as Auditor
13 actor "System\nAdministrator" as Admin
14 rectangle "E-Procurement Service" as EPS {
15   [Frontend Interfaces] as UI
16   [Backend/API Services] as API
17   database "Operational Database" as DB
18   [Fabric Client SDK] as SDK
19 }
20
21 cloud "Hyperledger Fabric Network" as HLF {
22   [Certificate Authority / MSP] as CA
23   [Endorsing Peer Nodes] as Peers
24   [Ordering Service] as Orderer
25   database "Channel Ledger" as Ledger
26   [Procurement Chaincode] as CC
27 }
28
29 Requester --> UI : create requisition\nconfirm receipt
30 Approver --> UI : approve/reject
31 Procurement --> UI : source, PO, contract
32 Supplier --> UI : register, quote, invoice
33 Finance --> UI : verify invoice\napprove payment
34 Auditor --> UI : review audit trail
35 Admin --> UI : configure roles/rules
36 UI --> API : REST/GraphQL requests
37 API --> DB : CRUD business records
38 API --> SDK : submit/evaluate transaction
39 SDK --> CA : identity enrollment
40 SDK --> Peers : proposal endorsement
41 Peers --> CC : validate asset rules
42 Peers --> Ledger : commit state
43 SDK --> Orderer : endorsed transaction
44 Orderer --> Peers : ordered block
45 @enduml
```

Listing 1. Overall actor-system interaction context

Overall high-level use case diagram

```
1 @startuml
2 ' Diagram: High-level use case model
3 ' Purpose: Capture externally visible functions of the e-
4   procurement service.
5 title High-Level E-Procurement Use Case Diagram
6 left to right direction
7 actor Requester
8 actor Approver
9 actor "Procurement Officer" as PO
10 actor Supplier
11 actor "Finance/AP Officer" as AP
12 actor Auditor
13 actor Admin
14 rectangle "E-Procurement System with Fabric Audit Layer" {
15   usecase "Register Vendor" as UC1
16   usecase "Create Requisition" as UC2
17   usecase "Approve Request" as UC3
18   usecase "Create RFO" as UC4
19   usecase "Submit Quotation" as UC5
```

```
20   usecase "Evaluate Quotation" as UC6
21   usecase "Issue Purchase Order" as UC7
22   usecase "Manage Contract" as UC8
23   usecase "Confirm Delivery" as UC9
24   usecase "Submit Invoice" as UC10
25   usecase "Verify Invoice" as UC11
26   usecase "Approve Payment" as UC12
27   usecase "Audit Compliance" as UC13
28   usecase "Resolve Exception" as UC14
29   usecase "Record Ledger Event" as UC15
30 }
```

```
31 Supplier --> UC1
32 Requester --> UC2
33 Approver --> UC3
34 PO --> UC4
35 Supplier --> UC5
36 PO --> UC6
37 PO --> UC7
38 PO --> UC8
39 Requester --> UC9
40 Supplier --> UC10
41 AP --> UC11
42 AP --> UC12
43 Auditor --> UC13
44 PO --> UC14
45 Admin --> UC15
46 UC2 .-> UC15 : <<include>>
47 UC3 .-> UC15 : <<include>>
48 UC7 .-> UC15 : <<include>>
49 UC11 .-> UC15 : <<include>>
50 UC12 .-> UC15 : <<include>>
51 @enduml
```

Listing 2. Overall high-level use case diagram

Component architecture diagram

```
1 @startuml
2 ' Diagram: High-level component architecture
3 ' Purpose: Map procurement functions to frontend, backend,
4   database, and Fabric components.
5 title E-Procurement and Hyperledger Fabric Component
6   Architecture
7 skinparam componentStyle rectangle
8 package "Client Layer" {
9   [Requester UI]
10  [Supplier Portal]
11  [Manager Approval UI]
12  [AP Workbench]
13  [Audit Dashboard]
14 }
15 package "Service Layer" {
16   [VendorService]
17   [RequisitionService]
18   [SourcingService]
19   [PurchaseOrderService]
20   [ContractService]
21   [ReceivingService]
22   [InvoiceService]
23   [PaymentService]
24   [AuditService]
25   [Fabric Gateway Adapter]
26 }
27 database "Operational SQL/NoSQL DB" as DB
28 folder "Object Store\nDocuments and Attachments" as Store
29 cloud "Hyperledger Fabric" {
30   [Fabric CA/MSP]
31   [Gateway / Client SDK]
32   [Procurement Chaincode]
33   [Endorsing Peers]
34   [Ordering Service]
35   database "Channel Ledger" as Ledger
36   database "Private Data Collections" as PDC
37 }
38 [Requester UI] --> [RequisitionService]
39 [Supplier Portal] --> [VendorService]
40 [Supplier Portal] --> [InvoiceService]
41 [Manager Approval UI] --> [RequisitionService]
42 [AP Workbench] --> [InvoiceService]
43 [Audit Dashboard] --> [AuditService]
44 [VendorService] --> DB
```

```

43 [InvoiceService] --> DB
44 [AuditService] --> DB
45 [ContractService] --> Store
46 [Fabric Gateway Adapter] --> [Gateway / Client SDK]
47 [Gateway / Client SDK] --> [Procurement Chaincode]
48 [Procurement Chaincode] --> [Endorsing Peers]
49 [Endorsing Peers] --> Ledger
50 [Endorsing Peers] --> PDC
51 [Gateway / Client SDK] --> [Ordering Service]
52 [Ordering Service] --> [Endorsing Peers]
53 @enduml

```

Listing 3. Component architecture diagram

BP01 Vendor Registration and Onboarding - Activity Diagram

```

1 @startuml
2 ' Diagram: BP01 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Vendor Registration and
  Onboarding.
4 title BP01 Activity Diagram - Vendor Registration and
  Onboarding
5 |Vendor/Supplier|
6 start
7 :New supplier wishes to transact or is invited by
  procurement;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for VendorProfile;
20 |Chaincode/Smart Contract|
21 :Execute RegisterVendor and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit VendorRegistered;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 4. BP01 Vendor Registration and Onboarding - Activity Diagram

BP01 Vendor Registration and Onboarding - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP01 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP01 Sequence Diagram - Vendor Registration and
  Onboarding
5 actor "Vendor/Supplier" as Actor
6 boundary "Supplier Portal: Registration Screen" as UI
7 control "VendorService API" as API

```

```

9 database "Vendor, VendorDocument, RiskCheck" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Vendor Registration and Onboarding
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(RegisterVendor, assetId, hash,
  status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit VendorRegistered event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 5. BP01 Vendor Registration and Onboarding - Sequence Diagram

BP01 Vendor Registration and Onboarding - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP01 state machine
3 ' Purpose: Show lifecycle states of the VendorProfile asset
  managed by backend workflow and chaincode.
4 title BP01 State Machine - VendorProfile
5 [*] --> Draft
6 Draft --> Submitted : next
7 Submitted --> UnderReview : next
8 UnderReview --> Approved : next
9 Approved --> Active : next
10 Active --> Suspended : suspended
11 Suspended --> Rejected : rejected
12 Rejected --> Archived : archived
13 Submitted --> Rejected : reject
14 Active --> Suspended : compliance hold
15 Suspended --> Active : reinstate
16 Archived --> [*]
17 @enduml

```

Listing 6. BP01 Vendor Registration and Onboarding - State Machine Diagram

BP02 Purchase Requisition Creation - Activity Diagram

```

1 @startuml
2 ' Diagram: BP02 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Purchase Requisition
  Creation.
4 title BP02 Activity Diagram - Purchase Requisition Creation
5 |Requesting Department|
6 start
7 :Business need for goods or services;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for PurchaseRequisition;
20 |Chaincode/Smart Contract|
21 :Execute CreateRequisition and validate lifecycle rules;

```

```

22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit RequisitionCreated;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 7. BP02 Purchase Requisition Creation - Activity Diagram

BP02 Purchase Requisition Creation - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP02 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
4 database, Fabric SDK, chaincode, peers, ordering
5 service, and ledger.
6 title BP02 Sequence Diagram - Purchase Requisition Creation
7 autonumber
8 actor "Requesting Department" as Actor
9 boundary "Requester Web/Mobile Requisition Form" as UI
10 control "RequisitionService API" as API
11 database "Requisition, RequisitionLine, BudgetReservation"
12 as DB
13 control "Hyperledger Fabric Client SDK" as SDK
14 control "Procurement Chaincode" as CC
15 participant "Endorsing Peer Nodes" as Peers
16 participant "Ordering Service" as Orderer
17 database "Channel Ledger / Audit Trail" as Ledger
18
19 Actor -> UI: Initiate Purchase Requisition Creation
20 UI -> API: Submit validated request payload
21 API -> API: Authorize identity and apply business rules
22 API -> DB: Store off-chain data and document references
23 DB --> API: Record identifier and content hash
24 API -> SDK: submitTransaction(CreateRequisition, assetId,
25 hash, status)
26 SDK -> Peers: Transaction proposal with client identity
27 Peers -> CC: Invoke chaincode validation
28 CC -> CC: Check asset state, role, and policy constraints
29 CC --> Peers: Read/write set and endorsement response
30 Peers --> SDK: Endorsed proposal responses
31 SDK -> Orderer: Broadcast endorsed transaction
32 Orderer -> Peers: Deliver ordered block
33 Peers -> Ledger: Validate endorsements and commit block
34 Ledger --> API: Emit RequisitionCreated event
35 API -> DB: Update status and event reference
36 API --> UI: Confirmation, next task, or exception message
37 UI --> Actor: Display outcome
38 @enduml

```

Listing 8. BP02 Purchase Requisition Creation - Sequence Diagram

BP02 Purchase Requisition Creation - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP02 state machine
3 ' Purpose: Show lifecycle states of the PurchaseRequisition
4 asset managed by backend workflow and chaincode.
5 title BP02 State Machine - PurchaseRequisition
6 [*] --> Draft
7 Draft --> Submitted : next
8 Submitted --> UnderValidation : next
9 UnderValidation --> PendingApproval : next
10 PendingApproval --> Returned : next

```

```

10 Returned --> Approved : next
11 Approved --> Rejected : rejected
12 Rejected --> Cancelled : cancelled
13 Submitted --> Rejected : reject
14 Returned --> Draft : revise
15 Cancelled --> [*]
16 @enduml

```

Listing 9. BP02 Purchase Requisition Creation - State Machine Diagram

BP03 Purchase Request Approval - Activity Diagram

```

1 @startuml
2 ' Diagram: BP03 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
4 ledger recording points for Purchase Request Approval.
5 title BP03 Activity Diagram - Purchase Request Approval
6 |Approver/Manager|
7 start
8 :Submitted requisition requires approval;
9 :Enter or review process data;
10 |Frontend Interface|
11 :Validate mandatory fields and role permissions;
12 if (Input complete?) then (yes)
13 |Backend/API Service|
14 :Create or update off-chain business record;
15 :Apply policy, budget, status, and segregation checks;
16 if (Business rules satisfied?) then (yes)
17 |Database Layer|
18 :Persist operational record and document references;
19 |Fabric Client SDK|
20 :Submit transaction for ApprovalDecision;
21 |Chaincode/Smart Contract|
22 :Execute RecordApproval and validate lifecycle rules;
23 |Peer Nodes|
24 :Endorse transaction proposal;
25 |Ordering Service|
26 :Order endorsed transaction into block;
27 |Ledger/Audit Trail|
28 :Commit state and emit ApprovalRecorded;
29 |Backend/API Service|
30 :Notify next actor and update dashboard;
31 stop
32 else (no)
33 |Backend/API Service|
34 :Create exception or return for correction;
35 |Ledger/Audit Trail|
36 :Record exception hash where material;
37 stop
38 endif
39 else (no)
40 |Frontend Interface|
41 :Show validation errors and keep draft;
42 stop
43 endif
44 @enduml

```

Listing 10. BP03 Purchase Request Approval - Activity Diagram

BP03 Purchase Request Approval - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP03 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
4 database, Fabric SDK, chaincode, peers, ordering
5 service, and ledger.
6 title BP03 Sequence Diagram - Purchase Request Approval
7 autonumber
8 actor "Approver/Manager" as Actor
9 boundary "Manager Approval Worklist" as UI
10 control "ApprovalWorkflowService API" as API
11 database "ApprovalTask, DelegationRule, PolicyCheck" as DB
12 control "Hyperledger Fabric Client SDK" as SDK
13 control "Procurement Chaincode" as CC
14 participant "Endorsing Peer Nodes" as Peers
15 participant "Ordering Service" as Orderer
16 database "Channel Ledger / Audit Trail" as Ledger
17
18 Actor -> UI: Initiate Purchase Request Approval
19 UI -> API: Submit validated request payload

```

```

18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(RecordApproval, assetId, hash,
    status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit ApprovalRecorded event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 11. BP03 Purchase Request Approval - Sequence Diagram

BP03 Purchase Request Approval - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP03 state machine
3 ' Purpose: Show lifecycle states of the ApprovalDecision
    asset managed by backend workflow and chaincode.
4 title BP03 State Machine - ApprovalDecision
5 [*] --> Pending
6 Pending --> InReview : next
7 InReview --> ClarificationRequested : next
8 ClarificationRequested --> Approved : next
9 Approved --> Rejected : rejected
10 Rejected --> Escalated : next
11 Escalated --> Expired : next
12 Expired --> [*]
13 @enduml

```

Listing 12. BP03 Purchase Request Approval - State Machine Diagram

BP04 Request for Quotation Preparation and Publication Activity Diagram

```

1 @startuml
2 ' Diagram: BP04 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
    ledger recording points for Request for Quotation
    Preparation and Publication.
4 title BP04 Activity Diagram - Request for Quotation
    Preparation and Publication
5 |Procurement Officer|
6 start
7 :Approved requisition requires competitive quotation or
    tendering;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for RFQ;
20 |Chaincode/Smart Contract|
21 :Execute CreateRFQ and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit RFQCreated;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;

```

```

34 |Ledger/Audit Trail|
    :Record exception hash where material;
    stop
  endif
else (no)
  |Frontend Interface|
    :Show validation errors and keep draft;
  stop
endif
@enduml

```

Listing 13. BP04 Request for Quotation Preparation and Publication - Activity Diagram

BP04 Request for Quotation Preparation and Publication - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP04 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
    database, Fabric SDK, chaincode, peers, ordering
    service, and ledger.
4 title BP04 Sequence Diagram - Request for Quotation
    Preparation and Publication
5 autonumber
6 actor "Procurement Officer" as Actor
7 boundary "Procurement RFQ Workspace" as UI
8 control "SourcingService API" as API
9 database "RFQ, RFQLine, SupplierInvitation" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Request for Quotation Preparation and
    Publication
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(CreateRFQ, assetId, hash,
    status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit RFQCreated event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 14. BP04 Request for Quotation Preparation and Publication - Sequence Diagram

BP04 Request for Quotation Preparation and Publication - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP04 state machine
3 ' Purpose: Show lifecycle states of the RFQ asset managed by
    backend workflow and chaincode.
4 title BP04 State Machine - RFQ
5 [*] --> Draft
6 Draft --> InternalReview : next
7 InternalReview --> Published : next
8 Published --> OpenForBids : next
9 OpenForBids --> Closed : closed
10 Closed --> Cancelled : cancelled
11 Cancelled --> Awarded : next
12 Awarded --> [*]
13 @enduml

```

Listing 15. BP04 Request for Quotation Preparation and Publication - State Machine Diagram

BP05 Supplier Quotation Submission - Activity Diagram

```
1 @startuml
2 ' Diagram: BP05 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Supplier Quotation
  Submission.
4 title BP05 Activity Diagram - Supplier Quotation Submission
5 |Vendor/Supplier|
6 start
7 :Supplier receives RFQ invitation or public tender access;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for Quotation;
20 |Chaincode/Smart Contract|
21 :Execute SubmitQuotation and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit QuotationSubmitted;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml
```

Listing 16. BP05 Supplier Quotation Submission - Activity Diagram

BP05 Supplier Quotation Submission - Sequence Diagram

```
1 @startuml
2 ' Diagram: BP05 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP05 Sequence Diagram - Supplier Quotation Submission
5 autonumber
6 actor "Vendor/Supplier" as Actor
7 boundary "Supplier Quotation Portal" as UI
8 control "QuotationService API" as API
9 database "Quotation, QuotationAttachment, BidReceipt" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Supplier Quotation Submission
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB -> API: Record identifier and content hash
21 API -> SDK: submitTransaction(SubmitQuotation, assetId, hash
  , status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC -> Peers: Read/write set and endorsement response
26 Peers -> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
```

```
Orderer -> Peers: Deliver ordered block
Peers -> Ledger: Validate endorsements and commit block
Ledger --> API: Emit QuotationSubmitted event
API -> DB: Update status and event reference
API --> UI: Confirmation, next task, or exception message
UI --> Actor: Display outcome
@enduml
```

Listing 17. BP05 Supplier Quotation Submission - Sequence Diagram

BP05 Supplier Quotation Submission - State Machine Diagram

```
1 @startuml
2 ' Diagram: BP05 state machine
3 ' Purpose: Show lifecycle states of the Quotation asset
  managed by backend workflow and chaincode.
4 title BP05 State Machine - Quotation
5 [*] --> Draft
6 Draft --> Submitted : next
7 Submitted --> Withdrawn : next
8 Withdrawn --> Opened : next
9 Opened --> UnderEvaluation : next
10 UnderEvaluation --> Accepted : next
11 Accepted --> Rejected : rejected
12 Rejected --> Expired : next
13 Submitted --> Rejected : reject
14 Expired --> [*]
15 @enduml
```

Listing 18. BP05 Supplier Quotation Submission - State Machine Diagram

BP06 Quotation Evaluation and Supplier Award - Activity Diagram

```
1 @startuml
2 ' Diagram: BP06 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Quotation Evaluation and
  Supplier Award.
4 title BP06 Activity Diagram - Quotation Evaluation and
  Supplier Award
5 |Procurement Officer|
6 start
7 :RFQ closes and quotations are available for evaluation;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for AwardDecision;
20 |Chaincode/Smart Contract|
21 :Execute RecordEvaluation and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit EvaluationRecorded;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml
```

```

42 endif
43 @enduml

```

Listing 19. BP06 Quotation Evaluation and Supplier Award - Activity Diagram

BP06 Quotation Evaluation and Supplier Award - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP06 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP06 Sequence Diagram - Quotation Evaluation and
  Supplier Award
5 autonumber
6 actor "Procurement Officer" as Actor
7 boundary "Evaluation Committee Dashboard" as UI
8 control "EvaluationService API" as API
9 database "EvaluationScore, AwardRecommendation,
  COIDeclaration" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Quotation Evaluation and Supplier
  Award
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(RecordEvaluation, assetId,
  hash, status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit EvaluationRecorded event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 20. BP06 Quotation Evaluation and Supplier Award - Sequence Diagram

BP06 Quotation Evaluation and Supplier Award - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP06 state machine
3 ' Purpose: Show lifecycle states of the AwardDecision asset
  managed by backend workflow and chaincode.
4 title BP06 State Machine - AwardDecision
5 [*] --> NotStarted
6 NotStarted --> Opened : next
7 Opened --> Scoring : next
8 Scoring --> Clarification : next
9 Clarification --> Recommended : next
10 Recommended --> Approved : next
11 Approved --> Rejected : rejected
12 Rejected --> Awarded : next
13 Awarded --> Protested : next
14 Protested --> [*]
15 @enduml

```

Listing 21. BP06 Quotation Evaluation and Supplier Award - State Machine Diagram

BP07 Purchase Order Generation and Supplier Acceptance - Activity Diagram

```

1 @startuml
2 ' Diagram: BP07 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Purchase Order Generation
  and Supplier Acceptance.
4 title BP07 Activity Diagram - Purchase Order Generation and
  Supplier Acceptance
5 |Procurement Officer|
6 start
7 :Approved award, catalog requisition, or contract release;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for PurchaseOrder;
20 |Chaincode/Smart Contract|
21 :Execute IssuePO and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit POIssued;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 22. BP07 Purchase Order Generation and Supplier Acceptance - Activity Diagram

BP07 Purchase Order Generation and Supplier Acceptance - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP07 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP07 Sequence Diagram - Purchase Order Generation and
  Supplier Acceptance
5 autonumber
6 actor "Procurement Officer" as Actor
7 boundary "PO Management Workspace" as UI
8 control "PurchaseOrderService API" as API
9 database "PurchaseOrder, POLine, ERPPosting" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Purchase Order Generation and Supplier
  Acceptance
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(IssuePO, assetId, hash, status
  )

```

```

22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit POIssued event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 23. BP07 Purchase Order Generation and Supplier Acceptance - Sequence Diagram

BP07 Purchase Order Generation and Supplier Acceptance - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP07 state machine
3 ' Purpose: Show lifecycle states of the PurchaseOrder asset
  managed by backend workflow and chaincode.
4 title BP07 State Machine - PurchaseOrder
5 [*] --> Draft
6 Draft --> Issued : next
7 Issued --> Acknowledged : next
8 Acknowledged --> ChangeRequested : next
9 ChangeRequested --> Amended : next
10 Amended --> PartiallyDelivered : next
11 PartiallyDelivered --> Delivered : next
12 Delivered --> Closed : closed
13 Closed --> Cancelled : cancelled
14 Cancelled --> [*]
15 @enduml

```

Listing 24. BP07 Purchase Order Generation and Supplier Acceptance - State Machine Diagram

BP08 Contract Creation and Approval - Activity Diagram

```

1 @startuml
2 ' Diagram: BP08 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Contract Creation and
  Approval.
4 title BP08 Activity Diagram - Contract Creation and Approval
5 |Procurement Officer|
6 start
7 :Award requires formal contract or framework agreement;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for Contract;
20 |Chaincode/Smart Contract|
21 :Execute CreateContract and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit ContractCreated;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif

```

```

38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 25. BP08 Contract Creation and Approval - Activity Diagram

BP08 Contract Creation and Approval - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP08 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP08 Sequence Diagram - Contract Creation and Approval
  autonumber
5 actor "Procurement Officer" as Actor
6 boundary "Contract Lifecycle Workspace" as UI
7 control "ContractService API" as API
8 database "Contract, ContractVersion, Obligation" as DB
9 control "Hyperledger Fabric Client SDK" as SDK
10 control "Procurement Chaincode" as CC
11 participant "Endorsing Peer Nodes" as Peers
12 participant "Ordering Service" as Orderer
13 database "Channel Ledger / Audit Trail" as Ledger
14
15 Actor -> UI: Initiate Contract Creation and Approval
16 UI -> API: Submit validated request payload
17 API -> API: Authorize identity and apply business rules
18 API -> DB: Store off-chain data and document references
19 DB --> API: Record identifier and content hash
20 API -> SDK: submitTransaction(CreateContract, assetId, hash,
  status)
21 SDK -> Peers: Transaction proposal with client identity
22 Peers -> CC: Invoke chaincode validation
23 CC -> CC: Check asset state, role, and policy constraints
24 CC --> Peers: Read/write set and endorsement response
25 Peers --> SDK: Endorsed proposal responses
26 SDK -> Orderer: Broadcast endorsed transaction
27 Orderer -> Peers: Deliver ordered block
28 Peers -> Ledger: Validate endorsements and commit block
29 Ledger --> API: Emit ContractCreated event
30 API -> DB: Update status and event reference
31 API --> UI: Confirmation, next task, or exception message
32 UI --> Actor: Display outcome
33 @enduml

```

Listing 26. BP08 Contract Creation and Approval - Sequence Diagram

BP08 Contract Creation and Approval - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP08 state machine
3 ' Purpose: Show lifecycle states of the Contract asset
  managed by backend workflow and chaincode.
4 title BP08 State Machine - Contract
5 [*] --> Draft
6 Draft --> LegalReview : next
7 LegalReview --> SupplierReview : next
8 SupplierReview --> Approved : next
9 Approved --> Active : next
10 Active --> Amended : next
11 Amended --> Suspended : suspended
12 Suspended --> Expired : next
13 Expired --> Terminated : next
14 Active --> Suspended : compliance hold
15 Suspended --> Active : reinstate
16 Terminated --> [*]
17 @enduml

```

Listing 27. BP08 Contract Creation and Approval - State Machine Diagram

BP09 Goods or Service Delivery Confirmation - Activity Diagram

```
1 @startuml
2 ' Diagram: BP09 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Goods or Service Delivery
  Confirmation.
4 title BP09 Activity Diagram - Goods or Service Delivery
  Confirmation
5 |Receiving Officer|
6 start
7 :Supplier delivers goods or completes service milestone;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for DeliveryReceipt;
20 |Chaincode/Smart Contract|
21 :Execute RecordDelivery and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit DeliveryRecorded;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml
```

Listing 28. BP09 Goods or Service Delivery Confirmation - Activity Diagrams

BP09 Goods or Service Delivery Confirmation - Sequence Diagram

```
1 @startuml
2 ' Diagram: BP09 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP09 Sequence Diagram - Goods or Service Delivery
  Confirmation
5 autonumber
6 actor "Receiving Officer" as Actor
7 boundary "Receiving Mobile/Web App" as UI
8 control "ReceivingService API" as API
9 database "GoodsReceipt, InspectionRecord, DeliveryAttachment"
  as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Goods or Service Delivery Confirmation
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB -> API: Record identifier and content hash
21 API -> SDK: submitTransaction(RecordDelivery, assetId, hash,
  status)
```

```
SDK -> Peers: Transaction proposal with client identity
Peers -> CC: Invoke chaincode validation
CC -> CC: Check asset state, role, and policy constraints
CC --> Peers: Read/write set and endorsement response
Peers --> SDK: Endorsed proposal responses
SDK -> Orderer: Broadcast endorsed transaction
Orderer -> Peers: Deliver ordered block
Peers -> Ledger: Validate endorsements and commit block
Ledger --> API: Emit DeliveryRecorded event
API -> DB: Update status and event reference
API --> UI: Confirmation, next task, or exception message
UI --> Actor: Display outcome
@enduml
```

Listing 29. BP09 Goods or Service Delivery Confirmation - Sequence Diagram

BP09 Goods or Service Delivery Confirmation - State Machine Diagram

```
1 @startuml
2 ' Diagram: BP09 state machine
3 ' Purpose: Show lifecycle states of the DeliveryReceipt
  asset managed by backend workflow and chaincode.
4 title BP09 State Machine - DeliveryReceipt
5 [*] --> AwaitingDelivery
6 AwaitingDelivery --> Delivered : next
7 Delivered --> UnderInspection : next
8 UnderInspection --> Accepted : next
9 Accepted --> PartiallyAccepted : next
10 PartiallyAccepted --> Rejected : rejected
11 Rejected --> Closed : closed
12 Closed --> [*]
13 @enduml
```

Listing 30. BP09 Goods or Service Delivery Confirmation - State Machine Diagram

BP10 Invoice Submission - Activity Diagram

```
1 @startuml
2 ' Diagram: BP10 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Invoice Submission.
4 title BP10 Activity Diagram - Invoice Submission
5 |Vendor/Supplier|
6 start
7 :Supplier has delivered goods/services and seeks payment;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for Invoice;
20 |Chaincode/Smart Contract|
21 :Execute SubmitInvoice and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit InvoiceSubmitted;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
```

```

41 stop
42 endif
43 @enduml

```

Listing 31. BP10 Invoice Submission - Activity Diagram

BP10 Invoice Submission - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP10 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP10 Sequence Diagram - Invoice Submission
5 autonumber
6 actor "Vendor/Supplier" as Actor
7 boundary "Supplier Invoice Portal" as UI
8 control "InvoiceService API" as API
9 database "Invoice, InvoiceLine, InvoiceAttachment" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Invoice Submission
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(SubmitInvoice, assetId, hash,
  status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit InvoiceSubmitted event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 32. BP10 Invoice Submission - Sequence Diagram

BP10 Invoice Submission - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP10 state machine
3 ' Purpose: Show lifecycle states of the Invoice asset
  managed by backend workflow and chaincode.
4 title BP10 State Machine - Invoice
5 [*] --> Draft
6 Draft --> Submitted : next
7 Submitted --> Validated : next
8 Validated --> DuplicateSuspect : next
9 DuplicateSuspect --> Matched : next
10 Matched --> Exception : next
11 Exception --> Approved : next
12 Approved --> Paid : next
13 Paid --> Cancelled : cancelled
14 Exception --> Approved : resolve
15 Cancelled --> [*]
16 @enduml

```

Listing 33. BP10 Invoice Submission - State Machine Diagram

BP11 Invoice Verification and Three-Way Matching - Activity Diagram

```

1 @startuml
2 ' Diagram: BP11 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Invoice Verification and
  Three-Way Matching.

```

```

4 title BP11 Activity Diagram - Invoice Verification and Three
  -Way Matching
5 |Finance Officer|
6 start
7 :Invoice is submitted and ready for validation;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for InvoiceMatch;
20 |Chaincode/Smart Contract|
21 :Execute VerifyInvoiceMatch and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit InvoiceMatched;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 34. BP11 Invoice Verification and Three-Way Matching - Activity Diagram

BP11 Invoice Verification and Three-Way Matching - Sequence Diagram

```

1 @startuml
2 ' Diagram: BP11 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP11 Sequence Diagram - Invoice Verification and Three
  -Way Matching
5 autonumber
6 actor "Finance Officer" as Actor
7 boundary "AP Invoice Verification Workbench" as UI
8 control "MatchingService API" as API
9 database "MatchResult, ExceptionCase, ToleranceRule" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Invoice Verification and Three-Way
  Matching
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(VerifyInvoiceMatch, assetId,
  hash, status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block

```

```

30 Ledger --> API: Emit InvoiceMatched event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml

```

Listing 35. BP11 Invoice Verification and Three-Way Matching - Sequence Diagram

BP11 Invoice Verification and Three-Way Matching - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP11 state machine
3 ' Purpose: Show lifecycle states of the InvoiceMatch asset
  managed by backend workflow and chaincode.
4 title BP11 State Machine - InvoiceMatch
5 [*] --> ReadyForMatch
6 ReadyForMatch --> Matching : next
7 Matching --> Matched : next
8 Matched --> Exception : next
9 Exception --> Clarification : next
10 Clarification --> Approved : next
11 Approved --> Rejected : rejected
12 Exception --> Approved : resolve
13 Rejected --> [*]
14 @enduml

```

Listing 36. BP11 Invoice Verification and Three-Way Matching - State Machine Diagram

BP12 Payment Approval and Payment Instruction - Activity Diagram

```

1 @startuml
2 ' Diagram: BP12 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Payment Approval and
  Payment Instruction.
4 title BP12 Activity Diagram - Payment Approval and Payment
  Instruction
5 |Finance Officer|
6 start
7 :Invoice is matched or exception approval is complete;
8 :Enter or review process data;
9 |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12 |Backend/API Service|
13 :Create or update off-chain business record;
14 :Apply policy, budget, status, and segregation checks;
15 if (Business rules satisfied?) then (yes)
16 |Database Layer|
17 :Persist operational record and document references;
18 |Fabric Client SDK|
19 :Submit transaction for PaymentApproval;
20 |Chaincode/Smart Contract|
21 :Execute ApprovePayment and validate lifecycle rules;
22 |Peer Nodes|
23 :Endorse transaction proposal;
24 |Ordering Service|
25 :Order endorsed transaction into block;
26 |Ledger/Audit Trail|
27 :Commit state and emit PaymentApproved;
28 |Backend/API Service|
29 :Notify next actor and update dashboard;
30 stop
31 else (no)
32 |Backend/API Service|
33 :Create exception or return for correction;
34 |Ledger/Audit Trail|
35 :Record exception hash where material;
36 stop
37 endif
38 else (no)
39 |Frontend Interface|
40 :Show validation errors and keep draft;
41 stop
42 endif
43 @enduml

```

Listing 37. BP12 Payment Approval and Payment Instruction - Activity Diagram

BP12 Payment Approval and Payment Instruction - Sequence Diagram

```

@startuml
' Diagram: BP12 sequence model
' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
title BP12 Sequence Diagram - Payment Approval and Payment
  Instruction
autonumber
actor "Finance Officer" as Actor
boundary "Finance Payment Approval Screen" as UI
control "PaymentService API" as API
database "PaymentApproval, PaymentInstruction,
  BankValidation" as DB
control "Hyperledger Fabric Client SDK" as SDK
control "Procurement Chaincode" as CC
participant "Endorsing Peer Nodes" as Peers
participant "Ordering Service" as Orderer
database "Channel Ledger / Audit Trail" as Ledger

Actor -> UI: Initiate Payment Approval and Payment
  Instruction
UI -> API: Submit validated request payload
API -> API: Authorize identity and apply business rules
API -> DB: Store off-chain data and document references
DB --> API: Record identifier and content hash
API -> SDK: submitTransaction(ApprovePayment, assetId, hash,
  status)
SDK -> Peers: Transaction proposal with client identity
Peers -> CC: Invoke chaincode validation
CC -> CC: Check asset state, role, and policy constraints
CC --> Peers: Read/write set and endorsement response
Peers --> SDK: Endorsed proposal responses
SDK -> Orderer: Broadcast endorsed transaction
Orderer -> Peers: Deliver ordered block
Peers -> Ledger: Validate endorsements and commit block
Ledger --> API: Emit PaymentApproved event
API -> DB: Update status and event reference
API --> UI: Confirmation, next task, or exception message
UI --> Actor: Display outcome
@enduml

```

Listing 38. BP12 Payment Approval and Payment Instruction - Sequence Diagram

BP12 Payment Approval and Payment Instruction - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP12 state machine
3 ' Purpose: Show lifecycle states of the PaymentApproval
  asset managed by backend workflow and chaincode.
4 title BP12 State Machine - PaymentApproval
5 [*] --> PendingApproval
6 PendingApproval --> Approved : next
7 Approved --> InstructionSent : next
8 InstructionSent --> Settled : next
9 Settled --> Rejected : rejected
10 Rejected --> OnHold : next
11 OnHold --> Cancelled : cancelled
12 Cancelled --> [*]
13 @enduml

```

Listing 39. BP12 Payment Approval and Payment Instruction - State Machine Diagram

BP13 Procurement Audit and Compliance Review - Activity Diagram

```

1 @startuml
2 ' Diagram: BP13 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Procurement Audit and
  Compliance Review.
4 title BP13 Activity Diagram - Procurement Audit and
  Compliance Review
5 |Auditor/Compliance Officer|
6 start

```

```

7  :Scheduled audit, risk alert, dispute, management review, or regulatory request;
8  :Enter or review process data;
9  |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12   |Backend/API Service|
13   :Create or update off-chain business record;
14   :Apply policy, budget, status, and segregation checks;
15   if (Business rules satisfied?) then (yes)
16     |Database Layer|
17     :Persist operational record and document references;
18     |Fabric Client SDK|
19     :Submit transaction for AuditRecord;
20     |Chaincode/Smart Contract|
21     :Execute CreateAuditRecord and validate lifecycle rules;
22     |Peer Nodes|
23     :Endorse transaction proposal;
24     |Ordering Service|
25     :Order endorsed transaction into block;
26     |Ledger/Audit Trail|
27     :Commit state and emit AuditRecordCreated;
28     |Backend/API Service|
29     :Notify next actor and update dashboard;
30     stop
31   else (no)
32     |Backend/API Service|
33     :Create exception or return for correction;
34     |Ledger/Audit Trail|
35     :Record exception hash where material;
36     stop
37   endif
38 else (no)
39   |Frontend Interface|
40   :Show validation errors and keep draft;
41   stop
42 endif
43 @enduml

```

Listing 40. BP13 Procurement Audit and Compliance Review - Activity Diagram

BP13 Procurement Audit and Compliance Review - Sequence Diagram

```

1  @startuml
2  ' Diagram: BP13 sequence model
3  ' Purpose: Show interactions among user, frontend, backend, database, Fabric SDK, chaincode, peers, ordering service, and ledger.
4  title BP13 Sequence Diagram - Procurement Audit and Compliance Review
5  autonumber
6  actor "Auditor/Compliance Officer" as Actor
7  boundary "Audit and Compliance Dashboard" as UI
8  control "AuditService API" as API
9  database "AuditRecord, AuditFinding, ControlTest" as DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Procurement Audit and Compliance Review
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(CreateAuditRecord, assetId, hash, status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit AuditRecordCreated event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message

```

```

UI --> Actor: Display outcome
@enduml

```

Listing 41. BP13 Procurement Audit and Compliance Review - Sequence Diagram

BP13 Procurement Audit and Compliance Review - State Machine Diagram

```

1  @startuml
2  ' Diagram: BP13 state machine
3  ' Purpose: Show lifecycle states of the AuditRecord asset managed by backend workflow and chaincode.
4  title BP13 State Machine - AuditRecord
5  [*] --> Planned
6  Planned --> EvidenceCollection : next
7  EvidenceCollection --> UnderReview : next
8  UnderReview --> FindingRaised : next
9  FindingRaised --> Remediation : next
10 Remediation --> Closed : closed
11 Closed --> Escalated : next
12 Escalated --> [*]
13 @enduml

```

Listing 42. BP13 Procurement Audit and Compliance Review - State Machine Diagram

BP14 Dispute or Exception Handling - Activity Diagram

```

1  @startuml
2  ' Diagram: BP14 activity model
3  ' Purpose: Visualize the behavioral flow, decisions, and ledger recording points for Dispute or Exception Handling.
4  title BP14 Activity Diagram - Dispute or Exception Handling
5  |Procurement Officer|
6  start
7  :Mismatch, rejected delivery, supplier protest, invoice exception, contract dispute, or policy violation;
8  :Enter or review process data;
9  |Frontend Interface|
10 :Validate mandatory fields and role permissions;
11 if (Input complete?) then (yes)
12   |Backend/API Service|
13   :Create or update off-chain business record;
14   :Apply policy, budget, status, and segregation checks;
15   if (Business rules satisfied?) then (yes)
16     |Database Layer|
17     :Persist operational record and document references;
18     |Fabric Client SDK|
19     :Submit transaction for DisputeCase;
20     |Chaincode/Smart Contract|
21     :Execute OpenDispute and validate lifecycle rules;
22     |Peer Nodes|
23     :Endorse transaction proposal;
24     |Ordering Service|
25     :Order endorsed transaction into block;
26     |Ledger/Audit Trail|
27     :Commit state and emit DisputeOpened;
28     |Backend/API Service|
29     :Notify next actor and update dashboard;
30     stop
31   else (no)
32     |Backend/API Service|
33     :Create exception or return for correction;
34     |Ledger/Audit Trail|
35     :Record exception hash where material;
36     stop
37   endif
38 else (no)
39   |Frontend Interface|
40   :Show validation errors and keep draft;
41   stop
42 endif
43 @enduml

```

Listing 43. BP14 Dispute or Exception Handling - Activity Diagram

BP14 Dispute or Exception Handling - Sequence Diagram

```
1 @startuml
2 ' Diagram: BP14 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP14 Sequence Diagram - Dispute or Exception Handling
5 autonumber
6 actor "Procurement Officer" as Actor
7 boundary "Exception/Dispute Case Workspace" as UI
8 control "ExceptionService API" as API
9 database "ExceptionCase, CaseMessage, CorrectiveAction" as
  DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Dispute or Exception Handling
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(OpenDispute, assetId, hash,
  status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit DisputeOpened event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml
```

Listing 44. BP14 Dispute or Exception Handling - Sequence Diagram

BP14 Dispute or Exception Handling - State Machine Diagram

```
1 @startuml
2 ' Diagram: BP14 state machine
3 ' Purpose: Show lifecycle states of the DisputeCase asset
  managed by backend workflow and chaincode.
4 title BP14 State Machine - DisputeCase
5 [*] --> Open
6 Open --> Assigned : next
7 Assigned --> UnderInvestigation : next
8 UnderInvestigation --> WaitingForSupplier : next
9 WaitingForSupplier --> Resolved : next
10 Resolved --> Rejected : rejected
11 Rejected --> Escalated : next
12 Escalated --> Closed : closed
13 Closed --> [*]
14 @enduml
```

Listing 45. BP14 Dispute or Exception Handling - State Machine Diagram

BP15 Supplier Performance and Contract Closeout - Activity Diagram

```
1 @startuml
2 ' Diagram: BP15 activity model
3 ' Purpose: Visualize the behavioral flow, decisions, and
  ledger recording points for Supplier Performance and
  Contract Closeout.
4 title BP15 Activity Diagram - Supplier Performance and
  Contract Closeout
5 |Procurement Officer|
6 start
7 :Delivery completion, contract milestone, periodic review,
  or contract expiry;
8 :Enter or review process data;
```

```
|Frontend Interface|
:Validate mandatory fields and role permissions;
if (Input complete?) then (yes)
  |Backend/API Service|
  :Create or update off-chain business record;
  :Apply policy, budget, status, and segregation checks;
  if (Business rules satisfied?) then (yes)
    |Database Layer|
    :Persist operational record and document references;
    |Fabric Client SDK|
    :Submit transaction for SupplierPerformance;
    |Chaincode/Smart Contract|
    :Execute RecordPerformance and validate lifecycle rules;
    |Peer Nodes|
    :Endorse transaction proposal;
    |Ordering Service|
    :Order endorsed transaction into block;
    |Ledger/Audit Trail|
    :Commit state and emit SupplierPerformanceRecorded;
    |Backend/API Service|
    :Notify next actor and update dashboard;
  stop
else (no)
  |Backend/API Service|
  :Create exception or return for correction;
  |Ledger/Audit Trail|
  :Record exception hash where material;
  stop
endif
else (no)
  |Frontend Interface|
  :Show validation errors and keep draft;
  stop
endif
@enduml
```

Listing 46. BP15 Supplier Performance and Contract Closeout - Activity Diagram

BP15 Supplier Performance and Contract Closeout - Sequence Diagram

```
1 @startuml
2 ' Diagram: BP15 sequence model
3 ' Purpose: Show interactions among user, frontend, backend,
  database, Fabric SDK, chaincode, peers, ordering
  service, and ledger.
4 title BP15 Sequence Diagram - Supplier Performance and
  Contract Closeout
5 autonumber
6 actor "Procurement Officer" as Actor
7 boundary "Supplier Performance Dashboard" as UI
8 control "PerformanceService API" as API
9 database "SupplierScorecard, KPIResult, ContractCloseout" as
  DB
10 control "Hyperledger Fabric Client SDK" as SDK
11 control "Procurement Chaincode" as CC
12 participant "Endorsing Peer Nodes" as Peers
13 participant "Ordering Service" as Orderer
14 database "Channel Ledger / Audit Trail" as Ledger
15
16 Actor -> UI: Initiate Supplier Performance and Contract
  Closeout
17 UI -> API: Submit validated request payload
18 API -> API: Authorize identity and apply business rules
19 API -> DB: Store off-chain data and document references
20 DB --> API: Record identifier and content hash
21 API -> SDK: submitTransaction(RecordPerformance, assetId,
  hash, status)
22 SDK -> Peers: Transaction proposal with client identity
23 Peers -> CC: Invoke chaincode validation
24 CC -> CC: Check asset state, role, and policy constraints
25 CC --> Peers: Read/write set and endorsement response
26 Peers --> SDK: Endorsed proposal responses
27 SDK -> Orderer: Broadcast endorsed transaction
28 Orderer -> Peers: Deliver ordered block
29 Peers -> Ledger: Validate endorsements and commit block
30 Ledger --> API: Emit SupplierPerformanceRecorded event
31 API -> DB: Update status and event reference
32 API --> UI: Confirmation, next task, or exception message
33 UI --> Actor: Display outcome
34 @enduml
```

Listing 47. BP15 Supplier Performance and Contract Closeout - Sequence Diagram

BP15 Supplier Performance and Contract Closeout - State Machine Diagram

```

1 @startuml
2 ' Diagram: BP15 state machine
3 ' Purpose: Show lifecycle states of the SupplierPerformance
4   asset managed by backend workflow and chaincode.
5 title BP15 State Machine - SupplierPerformance
6 [*] --> DataCollection
7 DataCollection --> Scored : next
8 Scored --> UnderReview : next
9 UnderReview --> Acknowledged : next
10 Acknowledged --> ImprovementPlan : next
11 ImprovementPlan --> Preferred : next
12 Preferred --> Suspended : suspended
13 Suspended --> Closed : closed
14 Closed --> [*]
15 @enduml

```

Listing 48. BP15 Supplier Performance and Contract Closeout - State Machine Diagram

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